

Design Proposal Outline: Sustainable Factory in Vietnam

Stellum Industries

Muhammad Hassan Umar

December 1, 2025

16,337 Words

Table of Contents

1.0	Executive Summary.....	7
2.0	Introduction.....	8
3.0	Project Background and Literature Review.....	10
3.1	Project Background.....	10
3.1	Sustainability Frameworks.....	11
3.1.1	UN SDGs	11
3.1.2	LEED and BREEAM	12
3.1.3	Circular Economy	12
3.1.4	United Nations Framework Convention on Climate Change (UNFCCC).....	12
3.2	Sustainable Factory Case Studies	12
4.0	Design Specifications and Objectives.....	14
4.1	Summary of Client Needs.....	14
4.2	Design Objectives	14
4.3	Technical, Social, Economic and Ethical Requirements.....	15
4.3.1	Technical Requirements.....	15
4.3.2	Social Requirements	15
4.3.3	Economic Requirements	16
4.3.4	Ethical Requirements.....	16
4.4	Constraints	16
1.0	Site and Location Analysis.....	17
5.1	Infrastructure.....	17
5.2	Global Integration	18
5.2.1	Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). 19	
5.2.2	European Union-Vietnam Free Trade Agreement (EVFTA)	19

5.3 Industrial Policy and Commitment to the Future.....	20
5.4 Du Long Industrial Park as the Optimal Hub	20
5.4.1 Energy and ESG Alignment.....	20
5.4.2 Site Characteristics.....	21
5.4.3 Logistical Efficiency.....	22
5.4.4 Political Certainty and Investment Protection	23
6.0 Stakeholder Analysis.....	23
7.0 Competitor and Industry Analysis.....	25
7.1 Direct Competitors.....	25
7.2 Indirect Competitors	25
8.0 Concept Development and Design Alternatives	27
8.1 Location	27
8.2 Energy System	28
8.3 Product Design.....	29
9.0 Final Design Concept and Architectural Layout.....	29
9.1 Final Product Design.....	30
9.2 Architectural Layout and Workflow Efficiency	31
9.3 Material Choices and Justification.....	35
9.4 Integration of Bioclimatic and Passive Design Principles	36
10.0 Engineering System's Design	36
10.1 Solar Photovoltaic Array Design.....	36
10.2 Water System Design	37
10.3 Waste Management System	38
10.4 Smart Factory Technologies and Industry 4.0.....	39
11.0 Machinery, Technology and Operation	39

11.1 Machinery and Sustainable Technology	40
11.1.1 Pre-Processing Equipment	40
11.1.2 Forming Equipment	40
11.1.3 Thermal and Finishing Equipment.....	40
11.2 Maintenance and Lifecycle Planning.....	41
11.2.1 Maintenance Strategy.....	41
12.0 Transportation and Logistical Analysis.....	41
12.1 Construction-Phase Logistics.....	42
12.2 Operational Logistics.....	42
12.3 Carbon Footprint Minimization	43
13.0 Sustainability.....	43
13.1 Environmental Sustainability.....	43
13.2 Social Sustainability.....	44
13.3 Economic Sustainability	44
14.0 Costing, Budgeting and Financial Feasibility.....	45
14.1 Preliminary Cost Estimate	45
14.2 Annual Fixed OPEX.....	46
14.2.1 Labor Costs	46
14.2.2 Energy Costs	46
14.2.3 Maintenance.....	46
14.2.4 Facility Costs	47
14.2.5 Packaging.....	47
14.2.6 Insurance and Compliance.....	47
14.2.7 Logistics.....	47
14.3 Profit and Loss Statement	48

14.4 Triple Bottom Line Matrix.....	50
15.0 Business Case and Strategic Feasibility Analysis	51
15.1 Feasibility Analysis	51
15.2 SWOT Analysis.....	52
15.3 Risk Analysis	53
15.4 Corporate Social Responsibility.....	54
16.0 Safety Engineering and Ethical Considerations.....	54
16.1 Safety Engineering in Design and Operation.....	54
16.2 Ethical Responsibilities.....	55
17.0 Expected Outcomes and Projected Impact.....	56
18.0 Conclusion	58
References.....	60
Appendices.....	64
Appendix 6.0.....	64
Appendix 9.....	64
Appendix 14.....	65

List of Figures and Tables

Figure 1: Rendered Design of Steel Lunchbox.....	30
Figure 2: Blueprints of the Factory. Dimensional Layout (Left). Detailed Interior (Right).	32
Figure 3: Zone A Blueprint	32
Figure 4: Zone B Blueprint.....	32
Figure 5: Zone C Blueprint.....	33
Figure 6: Zone D Blueprint.....	34
Figure 7: Zone E Blueprint	34
Figure 8: 3D Render of Factory in DLIP	35
Figure 9: Power-Interest Matrix.....	64
Figure 10: Machine Cost Breakdown	65
Table I: Identification of Possible Stakeholders.....	23
Table II: Stakeholder Influence Matrix.....	24
Table III: Stakeholder Expectations and Ethical Considerations [20]	24
Table IV: Porter's Five Forces Analysis (https://www.investopedia.com/terms/p/porter.asp)	26
Table V: Location Pugh Matrix.....	27
Table VI: Energy System Pugh Matrix	28
Table VII: Product Design Pugh Matrix	29
Table VIII: Preliminary Cost Estimate.....	45
Table IX: Labor Cost Breakdown	46
Table X: Facility Cost Breakdown.....	47
Table XI: Logistical Cost Breakdown.....	48
Table XII: Product Units and Revenue Table.....	48
Table XIII: Triple Bottom Line Matrix	50
Table XIV:Feasibility Assessment	51
Table XV: Risk Analysis Table	53

1.0 Executive Summary

Steel is an infinitely recyclable material uniquely positioned to replace single use plastics and less durable alternatives in the packaging production sector, presenting a significant opportunity to capture the growing eco-conscious lunch box market.

This proposal details the development and implementation of “DuraBox,” a line of premium stainless-steel lunch boxes. Stellum Industries is a reliable and high-efficiency manufacturing facility dedicated to the sustainable production of durable and reusable stainless-steel lunchboxes. This project is strategically sited in Du Long Industrial Park (DLIP) in the Ninh Thuan Province of Vietnam, a choice that leverages the region’s industrial infrastructure and commitment to sustainable actions regarding production and energy development.

Market research shows that 75% of consumers aged 25 to 45 are willing to pay a 20% premium for sustainable, high-quality alternatives to plastic. Thus, the fundamental purpose of this factory is to provide a viable, environmentally superior solution to global plastic waste and the high energy demands incurred by conventional container manufacturing. The projected total capital expenditure (CAPEX) is deliberately maintained below the client’s limit at \$9.50 million, confirming the project’s immediate financial feasibility.

The core economic and environmental strategy of the design hinges on the principle of infinite material circularity. Stainless steel (SS304 and SS316) is the material of choice because it can be recycled perpetually with minimal material loss and insignificant degradation of material strength and mechanical properties. This principle allows for a genuine closed-loop system, allowing the factory to produce, reprocess and upscale its own end-of-life products and manufacturing scrap, minimizing municipal solid waste and dependence on raw materials.

The factory is engineered as an advanced Industry 4.0 facility, optimizing utility and material consumption and workflow efficiency in operations. Sustainability is integrated through high-efficiency systems including a photovoltaic (PV) installation on the roof designated to produce 80% to 100% of the daily 350kW energy demand, supported by a reserve battery system for continuous operation. Energy management is further enhanced by Variable Speed Drive (VSD) compressors and Recuperative Heat Exchangers in the annealing furnace, both of which yield substantial energy savings.

The selection of Vietnam is validated by its extensive global integration, offering tax and tariff benefits and access to key markets as mandated in their Free Trade Agreements (FTA). This strategic positioning paired with explicit ethical labor commitments referencing regional Vietnamese minimum wage standards, ensures the facility aligns with stringent international Environmental, Social, and Governance (ESG) mandates. By demonstrating this successful convergence of profitability and resource efficiency, the project directly addresses UN Sustainable Development Goals 7, 9, 12, and 13.

2.0 Introduction

Sustainable industrial development is recognized as a fundamental pillar for economic growth across developing nations. As the global demand for consumer foods continues to rise, modern factories must be designed to achieve an essential balance between economic viability and environmental and social responsibility. Traditional manufacturing models frequently result in high energy consumption, substantial greenhouse gas emissions and significant waste generation. The global community's perception of the industrial landscape has led to a shift towards integrated systems utilizing renewable energy, closed-loop production cycles and smart logistics to minimize the ecological footprint.

The primary purpose of this report is to provide a comprehensive, feasible and innovative concept proposal for a production plant manufacturing durable, reusable steel lunch boxes in a developing economy. This project offers a tangible solution to the rising global concern over plastic container pollution, which can take years to decompose, and the high energy demands associated with alternatives like glass, which also result in increased transportation costs due to weight. The final design emphasizes durability, safety, and reusability, positioning the steel tiffin as a superior, eco-friendly alternative to single-use plastics. The ultimate intention is to develop a design that reduces pollution, optimizes energy use, and minimizes operational costs, resulting in a model that successfully demonstrates both long-term sustainability and profitability.

The shift towards sustainable industrialization is profoundly important for factories operating in developing economies as it provides a critical pathway to achieving balanced and long-term success. While industrial growth traditionally offered jobs and economic stability, it often resulted in environmental degradation, high energy consumption from fossil fuels, and

resource depletion. To overcome these issues, sustainable practices are now vital, allowing nations to compete globally without relying on outdated, contaminating technologies. According to United Nations in 2025, “Investing in sustainable industrialization is key to accelerating economic growth and creating the decent jobs necessary to break the cycle of poverty and hunger which plagues the LLDCs (Landlocked Developing Countries),” [1]. This modern approach, which integrates renewable energy, advanced resource optimization, and environmental conservation, ensures that industrial development creates both economic stability and decent jobs while simultaneously protecting natural resources and aligning with international climate action mandates.

The client, with a defined budget of \$10 million, expects a concept that is both profitable and sustainable. The design must propose sustainable and innovative solutions for setting up a production plant of sustainable goods in a developing country, ensuring profitability and proactivity with respect to the latest sustainability trends. Key expectations require the concept to address, at minimum, site and location analysis, transportation and logistics, architecture and materials, machinery and technologies, waste management, safety and ethical considerations, and detailed costing and sustainability analysis. The proposal must be specific and practical enough to demonstrate the feasibility of the project, including land acquisition and the commencement of operations.

The proposed facility is designed to fully align with key international sustainability goals, providing measurable contributions to sustainable industrial practices:

- SDG 7 (Affordable and Clean Energy): Achieved through the strategic integration of highly efficient photovoltaic (PV) systems, aiming for 80% to 100% operations to function on renewable energy.
- SDG 9 (Industry, Innovation, and Infrastructure): Demonstrated by utilizing advanced machinery and smart systems, integrated with the recycling process for huge mass production, which is essential for modernization.
- SDG 12 (Responsible Consumption and Production): Central to the design philosophy, focusing on the use of 100% recyclable stainless steel (SS304 and SS316 grades) to create a closed-loop system where end-of-life products are perpetually remanufactured.

- SDG 13 (Climate Action): Addressed by minimizing the factory's carbon footprint and encouraging the reduction of single-use plastics, supporting the client's goal of following sustainability trends

The scope of this report covers the entire development cycle of the factory, including site selection (Vietnam's Du Long Industrial Park), architectural layout, engineering systems (energy, water, waste), detailed product design, costing, and a thorough financial feasibility analysis. The report first defines objectives and subsequent research into significant factors. Though conceptual, fiscal, environmental and social analysis are performed, the results of which greatly impact the factory and its production.

A primary limitation is the strict client budget of \$10 million, which must cover all aspects of the investment, including land acquisition, construction, machinery, and renewable energy integration. The facility requires a land size of 7,000m².

Key assumptions include assumptions regarding natural variations that may occur, many minor fluctuations in weather are not considered to alter production. The equipment and systems must be selected to be durable, well-established and capable of operating efficiently within the context of a developing nation. It is crucial that the solutions align with the skills of the local workforce, account for the variability of infrastructure, and manage logistical challenges. Thus, it is assumed the workforce is capable and tailored to immediate integration into the factory's ecosystem. For calculations, it is assumed that all units produced are sold every year.

3.0 Project Background and Literature Review

3.1 Project Background

Industrialization in the developing world is essential for job creation and economic stability but often leads to environmental degradation. Factories today frequently use fossil fuels and conventional technology resulting in high energy consumption, environmental contamination and resource depletion [2]. Developing nations face this unfair competitive pressure of achieving rapid development while adopting sustainable practices, especially since most of the planet's resources have already been extensively utilized by developed countries. This constraint has led to the emergence of industries that aim for sustainable development through renewable energy

systems, advanced resource optimization, and environmental conservation strategies. The rapid development of smart systems and technology has finally allowed these methods to become feasible, enabling mass production to be profitable and sustainable simultaneously.

The choice of stainless steel (SS304 and SS316) is a strategic environmental decision. Studies on production processes that utilize reusable materials and recycling facilities have the potential to significantly minimize waste output. The replacement of durable, recyclable materials with single-use products and packages leads to large decreases in energy usage and environmental emissions. Stainless steel is the key material because the metal never loses strength and quality, even after recycling and reusing it several times. Creating secondary steel from scrap requires up to ten times less energy than the production of steel from virgin iron ore [4]. This robust end-of-life value ensures economic stability and delivers a reduction in environmental impact, including minimized mining requirements and lower carbon dioxide emissions.

3.1 Sustainability Frameworks

The factory's design is benchmarked against critical international frameworks to validate its commitment to global best practices.

3.1.1 UN SDGs

The project aligns explicitly with the following goals:

- SDG 7: Due to significant investment in renewable energy systems that seamlessly integrate with smart systems for maximum efficiency.
- SDG 9: Demonstrated by integrating advanced smart systems with steel recycling for huge mass production which can utilize almost any steel waste to remake the product.
- SDG 12: Achieved through a closed-loop production system where used products can be returned and remade supporting the concept emphasized by Coelho et al. [5].
- SDG 13: Demonstrated by the significant effort to reduce the facility's carbon footprint using existing technology within budget constraints

3.1.2 LEED and BREEAM

These standards, which promote resource efficiency and passive design, are relevant for the factory's architectural choices. The use of insulated metal panels and highly reflective roofing aligns with the goal of high thermal resistance and reduced solar heat absorption, which minimizes the energy demand for cooling over the building's 50-year lease.

3.1.3 Circular Economy

The Circular Economy philosophy mandates designing out waste and is a core strategic concept. The infinite recyclability of steel is the material foundation of this model, ensuring that the factory's product is an asset to the circular economy rather than an end-of-life burden. The design ensures components are easily separable from non-ferrous elements to maximize the purity and recovery rate of steel scrap.

3.1.4 United Nations Framework Convention on Climate Change (UNFCCC)

This project directly supports the UNFCCC's objectives by aligning with Vietnam's vision for a less carbonized and climate-resilient country by 2050. The large-scale deployment of solar power and the strategy to utilize future wind energy sources align the factory's operations with net zero commitments [6].

3.2 Sustainable Factory Case Studies

Case studies proved the economic and environmental feasibility of the design principles.

Unilever in Vietnam has demonstrated the possibility of achieving a net zero in internal operations within Vietnam since 2022 by transitioning to renewable energy and eliminating direct emissions [7].

Toyota in Thailand has successfully implemented advanced water recycling systems and hybrid power, providing a benchmark for mitigating high water usage, a problem present in existing steel industries [8].

The Renault-Morocco operation provides a highly relevant case study demonstrating how a major industrial player in a developing economy can successfully balance economic growth with environmental responsibility, directly addressing the principles of sustainable finance. Renault Morocco's transition strategy based on quantitative data analysis proves that significant

environmental milestones are achievable within the manufacturing sector. They were able to reduce carbon dioxide emissions by 30% per vehicle produced between 2010 and 2020, showcasing a proactive approach to sustainable manufacturing that aligns with global decarbonization efforts. Furthermore, their renewable green energy integration meant that the Tangier plant became the world's first carbon-neutral automotive factory, while the SOMACA plant leverages wind energy to meet its energy demands, underscoring the dedication to integrating renewable energy sources. The company actively seeks to minimize carbon dioxide related costs and has adapted its operations to increase the use of recycled and renewable sources.

This case demonstrates that sustainability is not a cost but rather the driver of competitive advantage and financial stability. Renault Morocco sustained profitability despite challenging market conditions, this confirms the ability to sustain profitability while integrating sustainable practices. The implementation of circular economy principles and responsible sourcing establishes their dedication to recycling and repurposing end-of-life vehicles, which promote sustainable manufacturing by minimizing waste and maximizing resource efficiency. Local integration and supply chain optimization allow the company to source 65% of objectives locally.

This case study validates core assumptions of our design proposal:

- It highlights the feasibility of efficiency targets as high as 30%
- Sustainability being used to produce a competitive edge
- It brings attention to the benefits of local sourcing
- Confirms the environmental and economic benefits of optimizing the supply chain [9].

Bifacial PV panels are solar panels capable of converting sunlight into electricity from both their front and rear sides. The front side captures direct irradiance, while the rear side captures albedo radiation, which is light reflected off the surface below, such as a white, highly reflective membrane on the factory roof. This is highly applicable to green roofs. Using bifacial panels over highly reflective and light-colored membranes maximizes reflective light capture, significantly increasing overall power production compared to conventional mono-facial panels. This increased energy production is a primary benefit that generates a greater amount of energy

in the same installed area. The roofing also generates causes space cooling and is incredibly suitable for dusty regions when aligned vertically. In places such as Vietnam where dust and soiling can otherwise reduce efficiency of solar panels, such panels may have long-lasting benefits [10].

Therefore, it may be considered that the inclusion of bifacial PV panels is a key technological feature intended to increase energy yield and efficiency in the Du Long Industrial Park.

4.0 Design Specifications and Objectives

4.1 Summary of Client Needs

The client has allocated a non-negotiable budget of \$10 million, which must cover all Capital Expenditure (CAPEX) including but not limited to land acquisition, construction, machinery and renewable energy systems. The primary expectation is to produce a factory that manufactures sustainable goods and is profitable in a developing country. The client further requires that the factory must align with modern sustainability technologies especially with reference to the UN Sustainable Development Goals (SDGs) and the outcomes from the UNFCCC. The proposal must be specific and practical, demonstrating the feasibility of commencing operations, covering key areas such as site and location, logistics, architecture, materials, waste management, safety, and ethical considerations.

4.2 Design Objectives

The project's success will be measured against specific, quantifiable, achievable, relevant, and time-bound (SMART) objectives, derived directly from the client's brief and the strategic choice of a steel lunchbox, closed-loop model.

The factory will reduce external energy dependence. It will achieve a minimum of 80% to 100% operational energy consumption from an integrated photovoltaic system. This is feasible because it is going back if feasible, given Vietnam's high solar potential and the strategic location in the Ninh Thuan province. This directly addresses SDG 7 (Affordable and Clean Energy) and the client's sustainability mandate. We believe this can be completed within 12 months of the factory's commencement.

The factory will further implement material circularity. We will ensure that the final product design permits 100% recyclability of the stainless-steel raw materials at its end-of-life. This is possible due to the inherent properties of stainless steel. This directly addresses SDG12, responsible consumption and production, and the factory's core closed-loop strategy. This can be integrated into the design and confirmed prior to the commencement of production.

The factory will also minimize operational waste. A near-zero manufacturing waste status by maximizing the recyclability of metal scrap and the reuse of water/cleaning agents through a closed-loop system. This is enabled by 100% recyclability of steel and advanced waste management systems in Vietnam. It supports cost efficiency, reduces operation expenditure, and again is addressing SDG 12. This can be sustainably maintained within the first 6 full months of operation.

Lastly, the factory will maintain a budget and financial viability. It will keep the total capital expenditure for the entire project below \$10 million. This is achievable as current cost estimates are around \$9.5 million. This is the single most important client constraint and therefore must be adhered to as strictly as possible. This can be measured at the time of signing final contracts.

4.3 Technical, Social, Economic and Ethical Requirements

The design must satisfy a multi-faceted set of requirements to be considered successful.

4.3.1 Technical Requirements

- The machinery must utilize Industry 4.0 technology to maximize efficiency, flexibility, and quality assurance.
- The production line must be capable of processing SS304 and SS316 grade steel sheets.
- The architectural layout must support a linear I-shaped production flow to eliminate transport waste and optimize workflow.

4.3.2 Social Requirements

- The factory must ensure full compliance with Vietnam's labor laws, specifically adhering to the regional minimum wage for the province and adhering to the 48-hour work week limit.

- Safe working conditions must be maintained throughout the provision, maintained through the provision of safety equipment and the use of air-conditioned transport and cooling units to mitigate the humid climate.
- The project must commit to corporate social responsibility (CSR) through ethical sourcing, community engagement, contributing to local skill development.

4.3.3 Economic Requirements

- The Factory must be positioned to leverage Vietnam's Free Trade Agreements and EVFTA to ensure competitive export pricing through preferential near zero tariffs.
- The design must ensure a strong return on investment and a viable payback period (to be calculated in Section 14).
- The chosen technologies must have an optimal balance between CAPEX and OPEX prioritizing life cycle cost savings.

4.3.4 Ethical Requirements

- All labor and sourcing practices must be managed extensively following the principles of the EA Code of Ethics (to be explicitly referenced in Section 16).
- The use of recycled materials must be ethically transparent, ensuring fair labor and sourcing throughout the supply chain (e.g. in the scrap steel collection process)

4.4 Constraints

There are several constraints when taking into consideration the production of a factory:

1. The total project's CAPEX is strictly capped at \$10 million.
2. The land area of 7,000 m² must be modulated within the Du Long industrial park.
3. The factory design must manage Vietnam's tropical and sunny climate which is advantageous for solar power but requires cooling solutions to protect workers and equipment.
4. Raw material sourcing is constrained by the need for specific grades of stainless steel (SS304/SS316) from regional suppliers, necessitating efficient logistics planning.

5. The design must adhere to national regulations governing industrial operations, safety standards, and environmental permitting within the DLIP.
6. It must also comply with the limitations associated with operating in a developing country, including training the local workforce and maintaining rigorous inspection of facilities.

5.0 Site and Location Analysis

The project location analysis begins with the selection of Vietnam as the chosen developing economy specifically focusing on the South-Central Coast region, specifically in the province of Ninh Thuan. This decision is strategically justified by various industrial, social, economic, and political considerations. The emergence of Vietnam as a high-tech manufacturing hub is substantiated by a surge in foreign direct investment (FDI), the dramatic rise of manufactured products and industrial modernization [11]. Vietnam offers a significant comparative advantage, particularly regarding its competitive costs and expanding global trade integration through various trade agreements.

5.1 Infrastructure

Vietnam's national push for infrastructural modernization is the critical enabler of its transition into a high-tech manufacturing hub. The national budget allocated approximately 6.3% of its GDP in 2021 to infrastructure investment which was confirmed to rise to approximately 7% in 2022, 90 of which was spent from the public budget [12]. The remaining 10% was funded by public-private partnerships used to scale up infrastructure upgrades. This is where FDI's start to take a role in the growth of the national infrastructure. Future trends show that this FDI percentage may grow up to 20% in the coming years [13].

The decision to choose Vietnam is directly supported by high-level national infrastructure targets, which address the competitive constraints that plague other Southeast Asian economies. The government issued "Decision No. 12/QĐ-TTg" dated January 03, 2025 on which promotes the development of the road system for the 2021-2030 period aiming for a total length of approximately 5000 kilometers by 2030 [14]. This commitment is vital for reducing logistics costs, which are historically high due to reliance on congested road freight. Projects like the

extended North-South Expressway directly connect key economic zones, ensuring efficient cargo flow from manufacturing clusters to major deep-water ports.

As such, maritime transport is paramount. Due to Vietnam's export-based economy, there are extensive national plans on upgrading international gateway ports, ensuring the capability to handle larger vessels and higher throughput. Vietnam's 251 ports including 45 seaports are a magnet of FDI with large scale investments from shipping lines and joint-venture port companies globally. The goal is to increase national seaport capacity by around 50% by 2030 which highlights the nation's global supply chain resilience.

Vietnam's digital backbone is highly advanced on account of their recent digital transformation. The effects of the pandemic ramped up the demand for digital services which has led to 100% telecom and 4G coverage in the nation. The rapid rollout of 5G in 2022 highlights the versatility of their robust "soft infrastructure," [13].

The national power strategy is governed by the Power Development Plan VIII (PDP8), which addresses the dual challenge of rapidly escalating industrial demand and the global push for sustainability. Vietnam has emerged as a global leader in solar and wind power deployment. The PDP8 intends to reach over 47% renewable energy by 2030, aligning the country with net-zero commitments and appealing to global corporations with strict ESG (Environmental, Social, and Governance) mandates further expanding business and trade within areas such as Europe [15].

5.2 Global Integration

Industrial and infrastructural strategy must be integrated with a larger legal and economic framework that guarantees the high standards and market access necessary for a high-tech manufacturing project. The Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) and the European Union-Vietnam Free Trade Agreement (EVFTA) are crucial elements of this framework.

Vietnam's comparative industrial advantage is secured not only by competitive costs and physical infrastructure plans but more importantly by its new-generation Free Trade Agreements (FTAs).

The CPTPP and EVFTA transcend traditional tariff-reduction deals; they function as a legal guarantee for foreign investors, signaling a commitment to a transparent, high-standard business environment essential for sophisticated, export-oriented manufacturing.

5.2.1 Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)

The CPTPP, involving nations across the Asia-Pacific and North America fundamentally alters the operational landscape in Vietnam. The agreement ensures Vietnam's manufacturing outputs gain "preferential tariff treatment," which in the consolidated CPTPP document is defined as "the customs duty rate applicable to an originating good, pursuant to each Party's Tariff Schedule." This solidifies Vietnam's access to preferential, almost zero-tariff access to global markets. This secures an external market for finished products and facilitates the efficient, cost-effective sourcing of high-quality intermediate goods from CPTPP member states.

The CPTPP further promotes investment protection and transparency. Through national treatment (i.e. treating firms from other nations the same as the local ones) and Most-Favoured-Nation Treatment which does not allow preferential treatment to member states (i.e, if favor is granted to one member nation, all member nations must be granted the same favor). This provides an essential layer of legal certainty that insulates long-term capital investment from regulatory or political risks [16].

5.2.2 European Union-Vietnam Free Trade Agreement (EVFTA)

The EVFTA provides a focused link to one of the world's largest and most discerning markets, the European Union. It eliminates nearly 99% of tariffs on goods traded between the EU and Vietnam, providing manufacturers with an undeniable competitive edge over rivals in non-FTA countries. Repaired and remanufactured goods will be duty free further promoting a closed loop style of manufacturing .

The EVFTA is expected to Vietnam's GDP by 4.6 percent and its exports to the EU by 42.7 percent by 2025. The EVFTA opens Vietnam's service sectors to EU companies at a level exceeding WTO commitment [11]. This commitment promotes greater competition, leading directly to the improved quality, efficiency, and lower cost of logistics and supply chain services that a high-tech manufacturing facility will rely on to export goods from Ninh Thuan.

In summary, the CPTPP and EVFTA serve as the institutional infrastructure that locks in Vietnam's commitment to a competitive, rules-based, and globally integrated economy, effectively mitigating investment risk and validating the choice of location for a high-value manufacturing operation.

5.3 Industrial Policy and Commitment to the Future

The current industrial strategy represents a break from historical governmental shortcomings, providing confidence in long-term stability and planning. For example, a 2001 paper named, *The Current Vietnamese Steel Industry, and Its Challenges* by Quoc Anh Ho. This paper states "One problem plaguing Vietnam's industrial development is the lack of an overall plan by the government despite its stated commitment to such development," [17].

The contrast is evident. This statement, authored in 2001, highlights the government's lack of industrial planning. Where once there was a lack of foresight, the present state represents proactivity, planning, and reliability. Vietnam is planning on the construction of over 221 new industrial zones and 98 industrial zones expecting expansion or adjustment by 2030 [18]. This massive, quantified expansion is the most compelling evidence that the government has successfully shifted from reactive development to meticulous industrial and infrastructural foresight. Furthermore, growth of this magnitude is a testament to proper civil and industrial management, ensuring that these new "Industrial Parks" are high-quality, well-maintained, and designed for modern, high-tech manufacturing requirements.

5.4 Du Long Industrial Park as the Optimal Hub

One such industrial park is Du Long Industrial Park (DLIP). The selection of Du Long Industrial Park in the Ninh Thuan Province of Vietnam is a strategic move that perfectly synthesizes all the national advantages into a single, high-efficiency location. DLIP is not merely a site, it is the physical embodiment of Vietnam's forward-looking industrial and trade strategy.

5.4.1 Energy and ESG Alignment

DLIP is specifically located in Ninh Thuan, the national center for solar and wind energy. This position directly addresses the national PDP8 plan for renewable energy, guaranteeing that the high-tech manufacturing facility will have reliable, priority access to clean power. DLIP is a

Certified Green Industrial Park, meaning its internal infrastructure including specialized utilities, robust waste management, and smart grids is purpose-built to meet the strict ESG mandates of global corporations, especially those exporting to the discerning EU market via the EVFTA. This capability provides a critical competitive advantage over facilities reliant on fossil fuels.

5.4.2 Site Characteristics

The Du Long Industrial Park located in the Tuan Bac District of the Ninh Tuan Province provides specialized functional infrastructure critical for the successful deployment of a high-technology, sustainable manufacturing facility.

The region's climate is a key factor supporting the factory's sustainability objectives. The province is located in Vietnam's South Central Coast region, characterized by a tropical monsoon climate with high heat and dryness, strong winds, and high evaporation rates. The average annual temperature ranges from 26 to 27 degrees Celsius. The region experiences two distinct seasons: a long dry season (December to August) and a shorter rainy season (typically starting in October and ending in December). Ninh Tuan is one of the driest provinces, receiving annual high sunshine hours, averaging from 2000 to 2500 hours a year, which is highly favorable for maximizing the output of the planned rooftop photovoltaic energy system. The area is considered semi-arid and is the driest region in Vietnam. The low annual rainfall (ranging from 700 mm to 800 mm) and high evaporation rate mean the site faces a risk of drought, emphasizing the need for the planned closed-loop water management system already integrated by the DLIP.

The site's characteristics are highly conducive to large-scale industrial development. DLIP is the largest industrial park currently operating in Ninh Tuong, spanning 407 hectares. This scale provides the necessary space for the 7,000 sqm facility and future expansion. The land within the industrial park is cleared and compensated, simplifying the process of land acquisition for the project. While the province features deep topography in some areas, the industrial park itself has complete facilities and infrastructure for operation, supporting the planned single-story high bay factory structure.

Water supply is secured through the park's central infrastructure complemented by internal recycling mechanisms. The Industrial Park has a designated, dedicated clean water supply station with a total capacity of 15,000 cubic meters day and night. This capacity ensures a reliable water source for the factory's operation, particularly for the cleaning and sterilization

processes. The park features a completed wastewater treatment plant that adheres to strict quality standards with a capacity of 12,000 cubic meters day and night. This central facility rigorously treats all incoming wastewater meeting environmental standards, which is vital for minimizing the factory's environmental impact.

Furthermore, DLIP is a strategic choice for high renewable energy integration. Energy is supplied from the national grid via a high-capacity 110/22 kV transformer station located within the industrial park. Ninh Thuan province is the national center for solar and wind energy. The industrial park is intentionally surrounded by wind and solar power projects [19]. Ninh Thuan is among the highest solar irradiance in Vietnam at around 5.2 kWh/m² a day [20]. This means a 1 MWp rooftop photovoltaic array could generate up to 1.5 GWh/year. It should be considered that DLIP is recognized as an Eco-industrial park actively planning to harness abundant wind and solar energy resources and connect them to the industrial grid. This is intended to reduce pressure on the national utility grid and meet industrial demands, supporting the factory's target of 80-100% renewable energy operations

5.4.3 Logistical Efficiency

DLIP's strategic placement ensures it fully capitalizes on the national infrastructure push. The park benefits from the Decision No. 12/QĐ-TTg road network extension (North-South Expressway), dramatically reducing inland logistics costs and ensuring rapid cargo movement. Due to Pomina and OSS, the two closest suppliers of sheet steel being around 350 kilometers away from DLIP. There is a logistical charge of 50 USD per ton of transported raw material. Upon completion, this charge can be reduced to almost 30 USD. It may be further considered due to DLIP's growth; a steel production factory may begin production within the industrial park leading to nonexistent logistical costs for the transportation of raw materials.

It was further considered that DLIP's proximity allows for streamlined access to deep-water ports, supporting the national goal of 50% increased seaport capacity. This efficient cargo flow is essential for leveraging the preferential, near-zero tariffs provided by the CPTPP and EVFTA, making the final landed cost of the exported product highly competitive in North American, Asian, and European markets.

5.4.4 Political Certainty and Investment Protection

DLIP's existence and continued support is proof of the government's shift from historical policy gaps to deliberate, quantified industrial planning. This commitment is reinforced by explicit governmental support and special incentives including preferential tax rates and expedited licensing. The DLIP itself provides tax holidays for up to 4 years with 50% tax reductions for up to the following 9 years. These regional benefits, combined with the comprehensive investment protections offered by the National Treatment and Most-Favored-Nation Treatment clauses within the CPTPP and EVFTA, provide an unparalleled layer of legal certainty that insulates long-term capital investment from political and regulatory risks.

With a total project area exceeding 407 hectares and access to a highly skilled local labor pool of over 3,000 staff, DLIP offers the necessary operational scale, sustainable foundation, and institutional support required to develop a profitable and environmentally conscious high-tech project.

6.0 Stakeholder Analysis

A comprehensive analysis of project stakeholders is essential for identifying potential supporters, managing areas of resistance, and ensuring the successful adaptation of the DuraBox initiative. This section details the key individuals, groups, and organizations, both internal and external, who will be affected by or have influence over the project. For each group, assessments have been made of their level of interest, their level of influence/power, and appropriate engagement strategy required to secure their support. The goal is to maximize support by prioritizing engagement efforts toward those groups with high influence and high interest in the project's outcomes.

The identification of stakeholders is performed first. This project involves a diverse range of stakeholders at local, national and global levels therefore people that are involved in or impacted are identified.

Table I: Identification of Possible Stakeholders

Category	Stakeholders
Client	Client (Investors), Management Team, Design Team
Execution	Design Team, Engineering Team, Factory Workers, Suppliers

Regulatory	Vietnamese Government (National), Du Long Industrial Park (Local)
External	Local Community, Consumers (Local and Global)
Environment	The Natural Environment, Water Sources

To structure the analysis, influence and interest of the key stakeholders can be mapped using a stakeholder influence matrix.

Table II: Stakeholder Influence Matrix

Stakeholder	Power	Interest	Quadrant	Management Strategy
The Client	High	High	Manage Closely	Satisfy Budget and Profitability
Factory Workers	Low	High	Keep Informed	Ensure fair wages, training and safety. Fulfill EA Ethics: Welfare
Vietnamese Government	High	Low	Keep Satisfied	Ensure compliance with trade agreements and industrial policy (PDP8)
DLIP	High	High	Manage Closely	Provide jobs, adhere to local regulations, promote development

The quadrant helps determine their alignment and the strategies that should be taken to successfully satisfy the parties involved. The matrix can be found in appendix 6.0.

The inclusion ethical responsibilities, promoting and highlighting competence and accountability.

Table III: Stakeholder Expectations and Ethical Considerations [20]

Stakeholder	Primary Expectations	Ethical Responsibilities
The Client	Maximize ROI, Adherence to the \$10 million budget, Alignment with the SDGs	Integrity: Ensuring transparency in costing and reporting. Honesty: Providing realistic assessments of feasibility and risk.
Factory Workers	Safe working conditions, fair wages, training, and development	Welfare: Prioritizing worker health, safety, and training above all else. Respect: Adhering strictly to labor laws (48-hour week, DLIP wages).
Local Community	Job creation, Minimal environmental disruption, Community engagement	Community: Ensuring positive contribution to social and economic growth.
The Environment	Reduction of carbon emissions, Minimal water consumption and waste	Environment: Utilizing the closed-loop system and renewable energy to be good stewards of the environment.

7.0 Competitor and Industry Analysis

The factory operates within a broader food storage and container market in Vietnam, competing directly against existing products of high-quality stainless goods and indirectly against dominant plastic and glass packaging sectors.

7.1 Direct Competitors

These are manufacturers producing steel products, including cookware, bento boxes, and industrial steel components, some of which already export from Vietnam.

The existing stainless-steel producers in Vietnam include companies like Goldsun Group, which specializes in kitchen and outdoor appliances and cookware. A company called G.a HOMEFAVOR, which produces stainless steel lunch boxes for the Vietnamese market, also currently holds a significant share in the local market. Firms like Tatonka, which markets steel food containers globally, are competitors in the same market.

Furthermore, global suppliers like Feemio offer high-quality, customizable stainless steel bento boxes to the Vietnamese market at wholesale prices, often leveraging factory direct models from China.

The Stellum factory gains an advantage not merely from product type but from its closed-loop production model and commitment to Industry 4.0 efficiency. Competitors, while offering stainless steel, rarely incorporate the full lifecycle 100% recyclability guarantee into their core operational strategy. The modern concern for the planet's well-being influences people, especially Generation Z, into making sustainability-first buying decisions according to research. Most members of Gen Z are considered tech-savvy and are accustomed to making informed purchasing decisions. As such individuals go and join the consumer market, they are an ideal set of people to consume our product [21].

7.2 Indirect Competitors

These are competitors that are capturing or have captured the market using materials that the stainless-steel lunch boxes seek to replace.

Firms like Du Tan Plastics are major manufacturers of food containers in Vietnam. They produce plastic/PET containers. The market is dominated by the convenience and low cost of plastic, which presents a high threat of substitution. Glass containers, produced by manufacturers like O-I BJC Vietnam (a leading glass container producer) supply the beverage and food industry. While glass is recyclable, its production is energy-intensive, and its heavier weight increases transportation costs and emissions compared to steel. Companies like Direct Eco Green focus on paper, wooden, and biodegradable packaging for the take-away sector, appealing to the same environmentally conscious segment.

Stellum Industries factories are still at an advantage. Though more expensive than plastic/PET containers, many global nations are moving towards bans on one-time use plastic bags and boxes. Furthermore, these PET boxes have been proven to cause long-lasting adverse health effects due to microplastics. Lastly, paper, wooden, and biodegradable packaging are not recyclable. Paper that is soiled during use or treated with chemicals is unable to be recycled in any form. Stellum Industries boxes are lightweight, durable, and entirely recyclable and aims to dominate the regional and global markets out of Vietnam.

The project benchmarks against global standards by focusing on energy intensity, targeting lower kWh per unit produced than both conventional steel manufacturing and energy-intensive glass production. Furthermore, Stellum Industries aimed for near-zero operational waste, vastly exceeding the performance of linear plastic and glass models. Ensuring all raw material, steel coil, and scrap sourcing aligns with the highest social and environmental standards, a key purchasing requirement for discerning international buyers. Lastly, by promoting a return system like the Pfand system in Germany. People will be incentivized into returning their end-of-life products or used products back to the company in return for money they deposited at the time of purchase. The factory can refurbish used products and reproduce end-of-life products for minimal costs. Further increasing sustainability and profitability.

Table IV: Porter's Five Forces Analysis (<https://www.investopedia.com/terms/p/porter.asp>)

Force	Intensity	Analysis and Rationale
Threat of New Entrants	Low	The industry requires high barriers to entry, specifically specialized deep-drawing servo press technology and high CAPEX (\$10M). Furthermore, new entrants lack the established logistical pathways and tariff advantages guaranteed by Vietnam's CPTPP and EVFTA.

Bargaining Power of Buyers	Medium to High	Buyers (retailers, institutional clients, direct consumers) have many substitutes (plastic, glass, imported steel). The buyer's power is mitigated by the factory's unique value proposition: superior quality, durability, and verifiable closed-loop sustainability.
Bargaining Power of Suppliers	Low	While the factory relies on regional suppliers for new steel coil (e.g., Pomina, OSS, Hoa Phat), the supplier power is low because the primary raw material input is steel scrap/recycled steel. The factory's 100% recyclability model internalizes the supply chain, reducing reliance on external steel mills.
Threat of Substitutes	High	The threat is high due to the overwhelming market presence of cheap plastic containers and the wide availability of durable glass alternatives. The factory overcomes this by offering a product with a superior environmental lifecycle, zero toxicity (BPA-free), and extreme durability compared to plastic or glass.
Industry Rivalry	Medium	Rivalry exists among domestic plastic and glass manufacturers, and globally competitive steel producers. Your factory minimizes direct rivalry by focusing on a specific niche (sustainable, closed-loop, export-focused steel boxes) and leveraging high operational efficiency from Industry 4.0 automation.

8.0 Concept Development and Design Alternatives

The final design for the Sustainable Factory was not selected arbitrarily but was chosen after formal evaluation of three distinct concepts, each addressing the core client mandate of profitability, sustainability, and deployment within a developing country with a strict budget of \$10 million. The core rationale was to maximize material circularity and export competitiveness while minimizing environmental impact.

8.1 Location

The location is one of the biggest factors in the production of a product within a factory. The appropriate location sets up a company for long-term success. The biggest factors included solar potential, raw material supply, ports/exports logistics, infrastructure and costs and future scalability. Three locations were chosen, and a simple Pugh matrix was produced.

Table V: Location Pugh Matrix

Figures of Merit	Factor	Du Long Industrial Park	Phu My II	Chau Dau Industrial Zone
Solar Potential	0.3	10	7	8

Raw Material Supply	0.1	7	10	9
Ports/Export Logistics	0.2	8	10	9
Infrastructure and Costs	0.3	9	8	6
Future Scalability	0.1	9	7	4
Total		8.9	8.2	7.3

Du Long Industrial Park scored highly due to its incomparable solar potential. Furthermore, due to the requirement to minimize raw material usage, the raw material supply is not as important for future production. Despite being relatively further from ports and export logistical areas, DLIP is close enough for easy transportation due to the lightweight nature of the product.

8.2 Energy System

To maintain the sustainable title, it is responsible of the factory to be completely self-sufficient and based on renewable energy sources. The two narrowed down for the Ninh Thuan province of Vietnam were solar power and wind power. However, research shows the difference between monofacial and bifacial solar panels which was thus considered through a Pugh matrix.

Table VI: Energy System Pugh Matrix

Figures of Merit	Factor	Monofacial PV	Bifacial PV	Wind Power
Load/Solar Output	0.5	8	10	5
Storage Requirements	0.1	8	8	10
Infrastructure and Costs	0.3	9	8	4
Maintenance	0.1	8	9	6
Total		8.3	9.1	5.3

The Pugh matrix makes it evident that photovoltaic cells are the ideal choice. Bifacial photovoltaic technology can harvest sun radiation from both sides of the panel. This offers a unique installation layout that reduces the required area for the photovoltaic system. This ability to harvest the sun energy from both sides offers efficiency enhancements up to 54% in ideal

circumstances. However, it must be noticed that photovoltaic cells have a lower peak power output. However, their benefit is not in their solar output, rather their ability to maintain such an intensive industrial load. In fact, initial calculations have shown that a mono-facial photovoltaic cell would only be able to compensate for 70-85% of the required power required by the factory. However, bifacial photovoltaic cells were able to handle 80-100% of the load [10].

8.3 Product Design

To ensure profitability, a product must be desirable. Therefore, certain design choices were made for both production and consumer benefits. These considerations were shown through a Pugh matrix.

Table VII: Product Design Pugh Matrix

Figures of Merit	Factor	Modular Tiffin	Sandwich Box	Sectional Box
Repeatability/Scalability	0.1	7	10	8
Design	0.1	7	9	10
Material Requirements	0.3	4	9	7
Industrial Requirements	0.3	5	10	8
Cost	0.2	4	9	6
Total		4.9	9.4	7.5

The Sandwich Box style is a simple, clean, and durable design which makes it the ideal choice for production. It has very low raw material requirements and industrial requirements. It simply requires a deep drawing process. As it is sheet steel, it will require a 400-ton press. However, such requirements would have been needed for all other options. A sandwich box can have chamfered edges, which allows for easy removal from the mold, thus increasing efficiency and parts produced per minute (PPM).

9.0 Final Design Concept and Architectural Layout

The architectural design of the manufacturing facility is not merely an enclosure for machinery but is an active component of the overall sustainability and operational strategy. It ensures. The structure itself supports energy efficiency, workflow optimization, and employee

well-being. This integrated architecture minimizes the factory's environmental footprint while maximizing long-term operational effectiveness.

9.1 Final Product Design



Figure 1: Rendered Design of Steel Lunchbox

The final product is a durable, reusable, and hygienically superior food storage container, specifically engineered for longevity and full material circularity.

The lunchbox is designed using precision deep drawing techniques, resulting in a seamless, smooth, flushed body and lid as depicted in the image. The box is manufactured exclusively from high-quality, food-grade SS304 or SS316 stainless steel. These materials are selected for their non-reactive, non-toxic properties, ensuring no leaching of contaminants into food, and their exceptional resistance to wear, tear, and corrosion.

The components exhibit a brushed, high-quality finish achieved through a precise manufacturing process involving techniques like electropolishing to ensure a smooth, hygienic surface that minimizes the potential for bacterial accumulation.

The body of the lunchbox is formed from a single piece of steel, eliminating seams and welds on the internal surface, thereby enhancing durability and ease of cleaning. The lid is designed to be leak-proof, incorporating a high-quality, non-toxic seal made of silicone that is easily removable for cleaning. This seal is engineered to create a tight form and friction fit, making the box suitable for transporting liquids as well as solids.

The product's design directly supports the project's core commitment to sustainability and ethical consumerism. The stainless steel is incredibly durable, ensuring the tiffin has a lifespan of decades, positioning it as an anti-consumerist alternative to products with built-in obsolescence. The product is designed to be 100% recyclable at its end of life. Its construction uses only steel and a simple separable silicone seal, maximizing the purity and recovery rate of the steel scrap for reprocessing into new lunchboxes, validating the factory's closed-loop production system.

The use of ultrasonic and steam sterilization methods in the production process ensures the final product is hygienic and ready for immediate use, meeting high quality standards. The box directly supports the reduction of single-use plastics and harmful plastic waste that contaminates landfills and oceans, offering an environmentally responsible choice for consumers.

9.2 Architectural Layout and Workflow Efficiency

The fundamental design principle is to establish a Linear, I-Shaped Production Flow, directly supporting the objective of eliminating transport waste. The building's architecture is a single-story, high-bay structure designed around the sequence of operations established in the process flow: Receiving, Forming, Thermal/Cleaning, Finishing, Shipping.

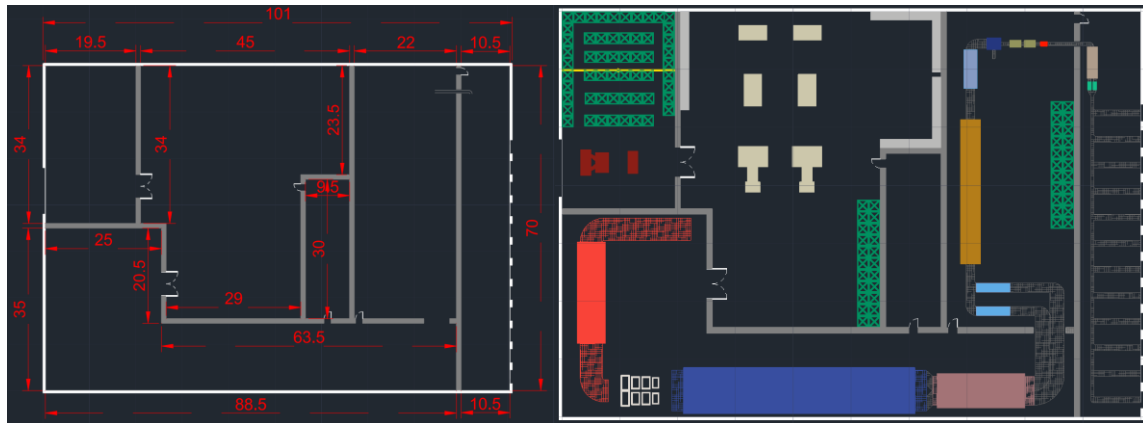


Figure 2: Blueprints of the Factory. Dimensional Layout (Left). Detailed Interior (Right).

The internal spatial design is zoned to optimize conditions for specific process steps. Zone A (Shown in Figure 2), the raw material receiving zone, accommodates the high-capacity overhead crane detailed in yellow and heavy-duty coil storage racks shown in green. The decoiler and straightener are shown in red with the sheet cutting shear/slitter shown in the same color on the right. The large bay doors on the left are for raw materials intake and are positioned to facilitate the efficient raw intake of raw steel coils and bulk materials. The large doors on the right allow for easy transportation of materials between zones.



Figure 3: Zone A Blueprint

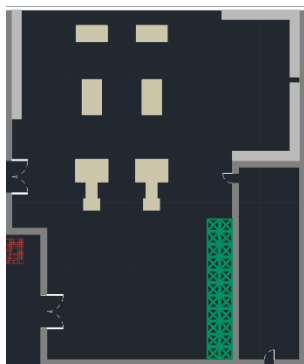


Figure 4: Zone B Blueprint

Zone B, C, and D are the core manufacturing zones. Zone B is responsible for high-impact operations, such as the heavy stamping of the servo presses, and is housed in a structure with isolated foundations to dampen vibration transfer. As shown in Figure 3, we notice it is relatively sparse with six machines responsible for all the deep drawing processes and a trimming press. Though not explicitly detailed, this area will have lubrication and blank feeder systems connected to all machinery. In gray, in the corners of the room, it can be noticed that there are workbenches supplied so that minor repairs and maintenance may be performed at any given moment.

Zone C is the most intensive part of the process. It involves the recuperative annealing furnace and water recycling system. Thus, it is specifically separated by reinforced moisture-

resistant walls to contain heat, steam, and chemical exposure, preventing cross-contamination with the final clean finishing areas. In Figure 5, we are able to notice a line of machinery. In red we notice the recuperative annealing furnace. Annealing is a heat treatment process which allows the steel to gain the properties we require. However, a lot of energy is usually lost. Therefore, by using a recuperative system in which the heat used is then stored and collected to continue heating the same machine, the energy lost through heat is significantly reduced. The white boxes between the annealing furnace and the electropolishing machine are ultrasonic, is an ultrasonic cleaning line with multi-stage immersions with filtration for any heat residue and material residue collected by the product to be removed. In blue, we notice the electro-polishing machine. Lastly, it goes through a drying tunnel which ensures the boxes are at an appropriate temperature to continue the process. Conveyor belts lead the product into zone D.

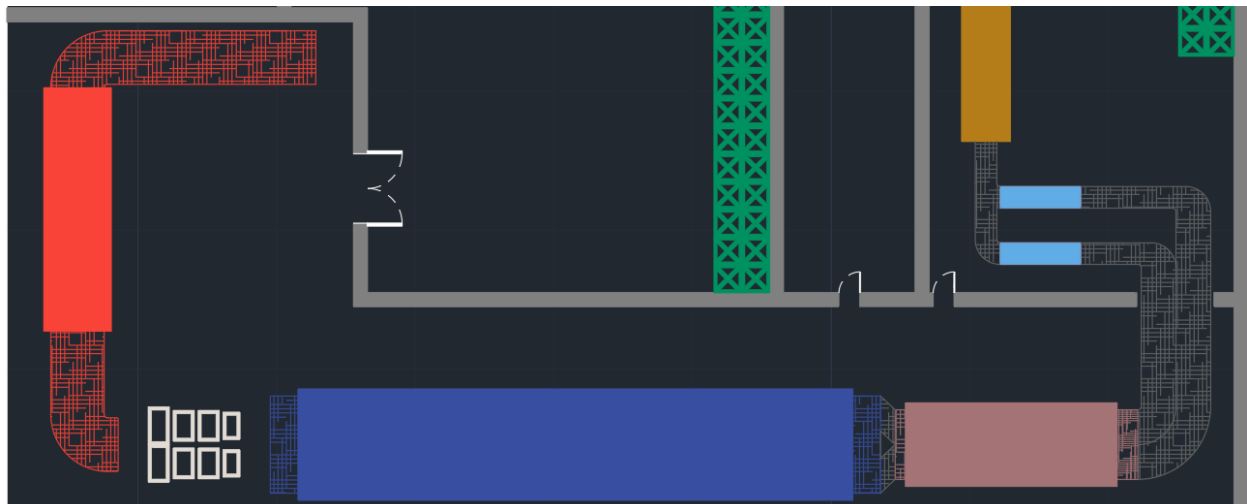


Figure 5: Zone C Blueprint

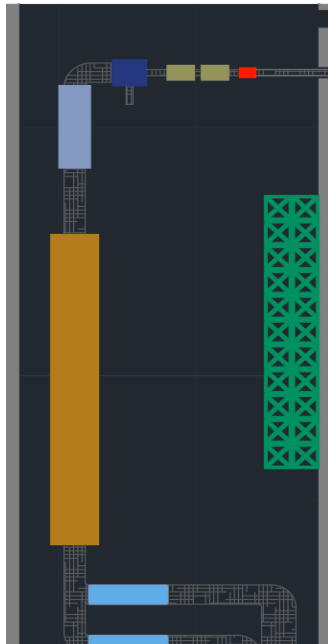


Figure 6: Zone D Blueprint

Zone D is the finishing and quality zone, housing the digital UV-LED printers shown in blue, and automated vision systems requiring stringent dust and temperature control. The electric curing oven is for the base and lacquer coats to produce the finishing on the product. The dark blue shows the Quality Inspection and Vision System, which is a high-resolution automated vision system to inspect in detail each product produced. Passed parts continue through the process. Rejected parts are discarded to be refurbished. High-speed CNC seamers are then used to set the final mechanical finishing and

ensure all edges are rounded and ready for use. In red, we notice the last step of the testing process: The leak and pressure tester using a non-destructive air leak testing system with high sensitivity to ensure that a closed product is completely airtight. It is located nearest to the final packing stage to minimize post-inspection handling.

Zone E is the final production zone. As the materials come in, they go through a final packaging line which is an automated cartoner for using recycled and compostable materials. The boxes are then sent through a barcode and data logging system in which all the barcodes can be tracked and traced in the time for delivery such that every product is recorded and accounted for. These are then conveyed into trucks which will export the products to the various port systems.

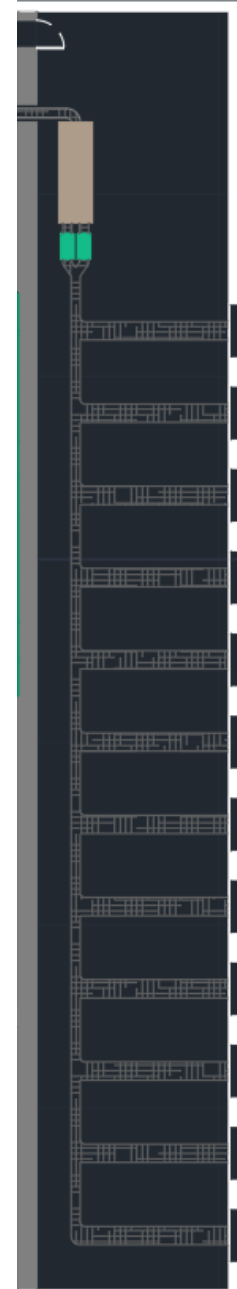


Figure 7: Zone E Blueprint

Zone F, though empty in the schematic, is the dedicated utility zone. It is compact and strategically isolated for the high demand equipment to centralize resources and minimize noise impact on the floor. It has access from every adjacent room to allow for rapid maintenance.

An important aspect of this design is the fact that the product after going through the electro-polishing tunnel is conveyed everywhere by conveyor belt represented by the grey with

the basket hatching. This is done purposefully so that there is minimal post-polishing handling. This reduces the chance of human error causing micro or macroscopic defects.



Figure 8: 3D Render of Factory in DLIP

9.3 Material Choices and Justification

The structural elements are selected not only for durability but to actively support the factory's energy efficiency objectives over its 50-year lease. The main structure of the factory is provided by DLIP as a shell, utilizing therefore these materials are the standard for construction in the region, making them both highly locally available and sustainable.

The walls are constructed using insulated metal panels (IMP). IMPs provide high thermal resistance compared to concrete, which in turn reduces heat transfer and lowers the demand for heating and cooling over the building's 50-year lease. They further simplify construction and site waste. The roofing will also be done using light-colored, highly reflective membranes which by having a high albedo surface reduce solar heat absorption, increasing the efficiency of the water-cooling system and reflecting the light back to the bi-facial photovoltaic cells.

In summary, the architectural design serves as the factory's primary control mechanism for energy and environmental management, ensuring that the physical structure actively reinforces the investment in high-efficiency equipment and the overarching commitment to sustainable manufacturing.

9.4 Integration of Bioclimatic and Passive Design Principles

Given Vietnam's tropical and sunny climate, the factory incorporates passive design principles to minimize reliance on active air conditioning and cooling units except when necessary for worker welfare, for example in staff areas or the quality to zone d or equipment protection.

The factory maximizes the use of natural light through strategically placed skylights and high-level windows to reduce the need for artificial lighting during daylight hours. This integrated with daylight sensors which automatically adjust the LED lighting output based on available natural light, delivering measurable energy savings of around 11% in office spaces and 28% in corridors [22].

The I-shaped layout is oriented to facilitate natural cross ventilation where possible, drawing in ambient air and expelling internal heat build-up, reducing reliance on mechanical cooling and improving worker comfort in non-critical zones.

The use of IMPs and reflective roofing manages the significant solar heat gain typical of the region, ensuring the interior climate remains stable and reducing energy demands on the air conditioning units that are necessary to mitigate the humid climate.

10.0 Engineering System's Design

The factory's energy system is designed to maximize renewable penetration and minimize reliance on the national grid, aligning with SDG 7 and Vietnam's PDP8. The system must meet the required energy demand of 350 kW daily and maintain critical operations overnight or during poor weather.

10.1 Solar Photovoltaic Array Design

The factory requires 350 kW of energy for its daily operations, assuming 8 hours of peak production, we find the following.

$$350 * 8 = 2800 \quad (1)$$

$$E_{peak} = 2800kWh \quad (2)$$

The energy required during off-peak hours must also be added. Considering a 0.8 MWh battery needing to be charged for other processes, we are able to find our daily energy requirement.

$$E_{daily} = 2800 + 800 \quad (3)$$

$$E_{daily} = 3600kWh \quad (4)$$

Assuming 5 hours of peaks hours and an 85% efficiency, the required installed capacity of the photovoltaic cells may be found.

$$P_{installed} = \frac{3600}{5 \cdot 0.85} \quad (5)$$

$$P_{installed} = 847.06kWp \quad (6)$$

Therefore, for full coverage and to meet the 80-100% renewable goal. The system is designed to have an installed capacity of 1000kWp using high efficiency photovoltaic bifacial panels exceeding the minimum requirement and generating surplus power for storage.

The design shows the solar panels at an 8.32° angle facing south as the ideal angle for the photovoltaic cells to be at maximum efficiency throughout the year [23]. Furthermore, this angle is at a 14.62% slope, which is ideal for Vietnam's tropical conditions, leading to dust and water to be drained properly off the roof.

The battery stored energy storage system is vital for energy reliability and maximum self-consumption. The required storage for basic functionality during non-production hours is 0.8 MWh. The system utilizes 500 kWh LiFePO₄ battery modules. This modular design provides flexibility for scaling and relies on LFP chemistry, which offers enhanced safety and longer cycle life compared to other lithium-ion chemistries. A 1 MW hybrid inverter/power control system manages the flow between PV array and battery, the national grid and factory load, ensuring uninterrupted power supply and optimized energy utilization.

10.2 Water System Design

This steel industry typically has a notorious water footprint. The design addresses this with the closed-loop philosophy, utilizing advanced purification and recycling to minimize freshwater consumption and environmental discharge.

Though Ninh Thuan is a dry region, rainwater harvesting provides a source of non-potable water for initial processes and utility needs. Large capacity cisterns are installed to collect runoff during the short rainy season and provide a reserve capacity to feed the grey water reuse system.

The water system targets near-zero discharges, ensuring the minimum wastewater generated can be treated by the Dulong Industrial Park Central Wastewater Treatment Plant.

The main water usage is for cleaning and sterilization. The design prioritizes low-water methods like ultrasonic cleaning systems and steam sterilization. The reverse osmosis and ion exchange system is the core technology. This system is installed downstream of the cleaning lines to capture filter and polish process water, making it suitable for reuse in non-critical applications such as the cooling towers and preliminary rinsing. This drastically reduces demand on the DLIP water supply.

10.3 Waste Management System

The waste management strategy is built upon the principle of circular economy, ensuring that waste is viewed as a resource. The foundation is the 100% recyclability of steel. Metal scraps generated during the forming process in Zone B are collected via an automated balling and compacting system and immediately routed back to the steel supply chain for remanufacturing, establishing a true closed-loop process.

The pre-built factory shell minimizes the waste produced during the facility setup. This allows for the complete removal of waste generation during external construction. Internally, waste may be produced during initial implementation. However, this is marginally less than the waste generated during construction.

A life cycle plan is established for the photovoltaic panels, which have an operation life of 20-35 years. End-of-life panels are committed to a specialized recycling technology that can achieve up to 96% recycling efficiency.

The use of an electrostatic micro-lubrication system in the forming process reduces lubricant consumption by up to 80% compared to conventional flood lubrication. Thus, minimizing the generation of hazardous liquid waste requires specialized disposal.

10.4 Smart Factory Technologies and Industry 4.0

“Industry 4.0, which is synonymous with smart manufacturing, is the realization of the digital transformation of the field, delivering real-time decision making, enhanced productivity, flexibility and agility to revolutionize the way companies manufacture, improve and distribute their products.” It represents the fourth industrial revolution which is characterized by the fusion of the industrial and digital worlds. It is built upon the implementation and integration of technologies such as the Internet of Things (IoT), cloud computing and analysis, Artificial Intelligence (AI), and machine learning [24].

The integration of IoT and smart systems is essential for achieving operational efficiency. Using centralized cloud-based systems integrated with sensors on critical machinery, such as the servo presses and furnaces, to perform real-time monitoring and predictive maintenance. This system detects anomalies and potential failures, preventing costly downtime and prolonging asset life.

Variable Speed Drive (VSD) compressors are used in place of conventional hydraulic presses. The VSDs are able to modulate motor speed based on fluctuating air demand and are managed by the smart system. This technology avoids inefficient unloaded states of fixed speed compressors, enabling typical energy reductions of 30% [25].

The Energy Management System integrated with the daylight sensors and hydro inverter monitors power generation from the PV system and directs power flow to the load of the battery. This real-time decision making ensures that power consumption is regulated based on renewable availability, maximizing efficiency and minimizing grid draw during peak hours.

11.0 Machinery, Technology and Operation

The operational strategy relies on advanced energy-efficient machine neutrals and specifically aligns with Industry 4.0 principles, maximizing precision and minimizing waste.

11.1 Machinery and Sustainable Technology

11.1.1 Pre-Processing Equipment

An automated leveling system is used. This system eliminates coil memory and strip shape defects directly reducing the potential for instability, wrinkling, and fractures during the deep drawing process. A strong foundation allows for a high-quality product to be produced.

An electrostatic micro-lubrication system replaces inefficient flood lubrication practices by applying a minimal electrically charged film of forming fluid. This precise deposition reduces lubrication consumption by up to 80% and simplifies cleaning processes, minimizing the generation of hazardous waste. Lubrication reduces wear and dislodged particles, thus improving the worker's welfare and increases the machinery's life expectancy.

11.1.2 Forming Equipment

The core of the production line consists of 400-ton electromechanical servo presses. This selection is a critical technological decision prioritizing enhanced capability and energy proportionality over conventional hydraulic systems. Unlike hydraulic presses that consume high power continuously, the servo system decouples ram force from speed, allowing for a low power state to be activated during idling. This capability is expected to reduce energy consumption by 25-30%.

The servo presses offer superior control over position, velocity, and force throughout the entire stroke. This kinematic profiling is vital for managing the material flow during deep drawing, which minimizes material thinning, internal stresses, and reduces failure modes like fracturing. Ultimately, this leads to less scrap.

11.1.3 Thermal and Finishing Equipment

Recuperative heat furnace is a continuous annealing furnace equipped with a recuperative heat exchanger (RHE). This technology captures latent heat energy from the exhaust flue gases and transfers it back to preheat the combustion air before it enters the burners. This heat recovery mechanism significantly increases thermal efficiency, effectively reducing net natural gas consumption, and lowering associated carbon dioxide emissions. [26]

Digital UV LED printers are employed for the decoration phase. This digital method eliminates the high setup costs and significant waste associated with traditional analog printing processes.

11.2 Maintenance and Lifecycle Planning

Life cycle planning informs the maintenance strategy and financial accounting for asset depreciation. Asset depreciation and end-of-life can be categorized into three different assets.

1. Long-term assets which are expected to have a life of 15 to 20 years are heavy-duty capital assets such as the servo presses and the annealing furnace. They are soon assigned a normal useful life (NUL) of 15 to 20 years. Their lifespan is extendable through rigorous maintenance protocols.
2. Medium-term assets are expected to have a life of 10 to 12 years. These components are subject to high duty cycles or rapid efficiency gains, such as the variable servo drive, compressors, and motor drives. These are assigned an annual life of 10 to 12 years.
3. Short-term assets are expected to have a life of 5-7 years. They are integrated digital and vision systems that have the shortest life due to rapid obsolescence of software and computing hardware.

11.2.1 Maintenance Strategy

The maintenance strategy shifts from reactive to predictive maintenance via the Industry 4.0 framework. IoT integration involves sensors connected to the ERP system that monitor vibration, temperature, and energy consumption in real-time. This allows maintenance teams to schedule interventions before failure occurs, maximizing uptime and overall equipment effectiveness (OEE). [27][28][29]

12.0 Transportation and Logistical Analysis

Logistical planning is a critical aspect of setting of the factory and sustaining daily operations. Proper management ensures timely transportation, decreases costs, and is fundamental to minimizing the overall carbon emissions of the supply chain.

12.1 Construction-Phase Logistics

The factory's construction logistics focus on minimizing freight distance and reducing the frequency of large-scale shipments. The heavy specialized machinery, such as the deep-draw servo presses and welding machines, will be sourced internationally from manufacturing hubs in countries like Japan and China due to the free trade agreement with these nations. These imported goods will arrive via Hai Phong port, one of Vietnam's industrial gateways, before being transported inland to the DLIP in Ninh Thuan province.

Locally sourced building materials, like concrete and structural steel, as well as the IMPs and reflective roofing, will be employed wherever feasible to minimize trade distance and benefit local suppliers. This strategy includes consolidating shipments into bulk delivery shipments to reduce transfer frequency and overall fuel use. This coordinated approach is projected to lower total logistical emissions by 15% to 25% [30].

12.2 Operational Logistics

Operation logistics cover the inbound and outbound flow of raw materials and the outbound distribution of the finished product, both domestically and for export.

Coils of raw stainless steel (SS304/SS316) will be supplied from regional Vietnamese steel producers such as Pomina and OSS, located approximately 300 to 350 km away in the Ba Ria-Vung Tau area. The government's investment in infrastructure, specifically the extended North-South Expressway, is crucial. This is expected to significantly reduce logistic costs inland. The current logistical charge of \$50 per ton of raw material is projected to decrease to approximately \$30 per ton upon the completion of the road network improvements.

Distribution routes satisfy the outbound logistical requirements. Finished lunch boxes will be delivered to major domestic population and commercial centers, including Ho Chi Minh City, Da Nang and Hanoi, using low-emission trucks. For export distribution, the primary port for international export will be Cam Ranh Port. Situated approximately 30 km from DLIP, it provides streamlined access to deep-water shipping lanes. Products will be shipped to neighboring nations and international markets leveraged by the EVFTA and CPTPP agreements.

12.3 Carbon Footprint Minimization

The logistics chain is designed to minimize its carbon footprint through strategic material choices and optimized transportation methods. Sea transport is the preferred method for international exports as it releases significantly less carbon dioxide than air transportation, ensuring the factory's global distribution remains low carbon. The domestic delivery fleet will gradually transition to electric vehicles as Vietnam's changing infrastructure evolves, further reducing greenhouse gas emissions. GPS scheduling and advanced routing systems will be utilized to plan delivery routes, reducing idle time and fuel waste.

A key innovation is the implementation of reverse logistics. Transportation timings will be planned so that delivery vehicles can collect used end-of-life boxes while returning from domestic distribution runs. This practice drastically reduces empty return trips, lowers fuel consumption, and is critical for feeding the factory's core closed-loop recycling process.

Packaging materials used for export will be recyclable and reusable, eliminating the use of any single-use plastics in the packaging process. The box's durability will be sufficient to protect its geometry during transportation.

These arrangements ensure that logistics are efficient, environmentally sustainable, and align with Vietnam's national vision of green transportation and low carbon emissions.

13.0 Sustainability

13.1 Environmental Sustainability

Stellum Industries aggressively mitigates its carbon footprint by focusing on energy sources and process optimization. The 1000 kW photovoltaic array supported by an 800 kWh battery energy storage system is designed to generate more than the required daily energy demand. This system aims to power 80-100% of factory operations, with clean energy providing direct quantifiable offset to solar carbon dioxide emissions that would otherwise be generated by grid power.

Manufacturing new steel from recycled scrap requires up to 10 times less energy than producing primary steel from virgin's iron ore. This material choice and process is the single

biggest factor in long-term carbon footprint minimization for the product life cycle. Furthermore, the adaptation of high-precision servo electromechanical servo presses reduces energy intensity by 25 to 30 percent, and the use of variable speed drive compressors cut power consumptions by up to 30 percent by eliminating inefficient unloaded states [25]. Recuperative heat exchanger on the annealing furnace increases thermal efficiency and reduces net natural gas consumption.

Waste circularity means that metal scraps generated during manufacturing are recycled in a closed loop system, diverting waste from landfills. The durable stainless-steel product itself has 100% recyclable and incredibly durable properties, directly reducing the volume of single-use plastic waste in the environment. The system prioritizes water efficiency through low-water processes like ultrasonic cleaning and steam sterilization. The use of reverse osmosis and ion exchange system allows for water processes to be filtered and reused, dramatically reducing demand on municipal supply and minimizing wastewater discharge.

13.2 Social Sustainability

The project's social strategy is centered on creating high-quality employment, ensuring worker welfare and contributing positively to the local community in the Ninh Thuan province.

The factory strictly adheres to the Vietnamese labor code, ensuring maximum working hours do not exceed 48 hours per week, all workers receive at least the minimum wage for the region and the project prioritizes operational safety by providing all workers with mandated safety equipment and PPE.

This investment will generate long-term job opportunities for the local population. This industry can benefit the local area by building trained and skilled workers for manufacturing and welding in many other areas. The project will inspire people to choose reusable boxes over plastic ones, helping to create responsible habits among consumers.

13.3 Economic Sustainability

The long-term economic viability is based on maximizing operational efficiency, leveraging global trade integration and realizing long-term cost savings from sustainability investments

The efficiency gains from energy systems result in long-term cost savings. The total energy consumption of the factories aims to be reduced by a minimum of 30%. The closed loop system converts production waste from a disposable cost into a reusable asset, significantly reducing raw material input costs over a long time. The industry 4.0 system enables predictive maintenance, and the precision servo press operation reduces wear and tooling, leading to less downtime, lower repair costs, and prolonged asset life.

Investments will contribute to national income by benefiting from exporting steel products to global markets. The preferential tariffs offered by the EVFTA and CPTPP provide a competitive edge. Furthermore, special incentives from the Dulong Industrial Park, including tax holidays for up to four years and 50% tax reduction for the following nine years, provide a substantial economic advantage and insulate the long-term capital investment from political and regulatory risks [24].

14.0 Costing, Budgeting and Financial Feasibility

14.1 Preliminary Cost Estimate

The total initial capital expenditure (CAPEX) for the Stellum Industries factory is estimated at \$9,500,000, successfully operating within the client's \$10M limit. The largest component of the budget is allocated to high-efficiency specialized machinery and equipment costing \$3,489,000, and the energy system which is \$1,230,000. These are investments critical to guaranteeing both sustainability and long-term operational cost savings.

The cost justification is anchored in the selection of technologies that reduce operational expenditure (OPEX). Although more expensive than traditional hydraulic presses, servo presses offer 25-30% energy savings and higher precision, which reduces material scrap rates. for the moment, the investment in the 1000 kWp solar array and battery storage directly hedges against future energy price volatility and provides an immediate reduction in utility bills.

Table VIII: Preliminary Cost Estimate

Category	Note	Cost (USD)
Land Acquisition	\$195/sqm (10,000m ²)	1,950,000
Shell Factory Purchase	110USD/sqm (7000 m ²)	490,000
Interior Construction	Walls etc.	770,000

Machinery and Equipment	See Appendix for Breakdown	3,489,000
Energy System	See Appendix for Breakdown	1,230,000
Raw Material Inventory	Incl. Transportation	160,000
Site Prep	Permits and Grid integration	350,000
Working Capital	Wages etc.	300,000
Contingency		750,000
	Total	9,500,000

14.2 Annual Fixed OPEX

Assuming the factory produces 600,000 new units per year and 300,000 refurbished units per year, the total annual OPEX is roughly equal to \$2.55 million a year. This value is broken down below.

14.2.1 Labor Costs

Table IX: Labor Cost Breakdown

Role	Number	Monthly Salary	Annual Cost
Line Workers	55	\$380	\$250,800
Skilled Technicians	8	\$700	\$67,200
Engineers	4	\$1,200	\$108,000
Logistics Staff	8	\$500	\$48,000
Ground Staff	6	\$350	\$25,200
Total	81	\$3130	\$556,800

14.2.2 Energy Costs

The factory has renewable on-site power generation. However, grid balancing, peak-demand supplies and battery maintenance are unavoidable, therefore, \$25,000 can be allocated annually for OPEX energy costs.

14.2.3 Maintenance

Deep drawing dies and tooling upkeep is the biggest fixed OPEX for metal factors. These include:

- Repair to dies
- Replacement drawing rings
- Lubricants

- Polish
- Bearing bolts
- Belts
- Other mechanical parts

The maintenance cost is capped at 4% of the CAPEX machinery cost. Therefore, a value of \$140,000 per year is obtained.

14.2.4 Facility Costs

Even if the land is purchased within DLIP, certain taxes, permits, and fees still apply for the duration of the lease.

Table X: Facility Cost Breakdown

Cost	Annual (\$)
Land Tax	40,000
Water	24,000
Wastewater Treatment	15,000
DLIP Service Fee	22,000
Security and Maintenance	18,000
Total	119,000

14.2.5 Packaging

All products need packaging, this includes cartons, dividers, and labels which amount to \$90,000.

14.2.6 Insurance and Compliance

To adhere to local codes, administrative costs will apply, including factory insurance, workers' comp, quality ISO audits, office supplies, as well as IT and software. This amounts to \$75,000 per year.

14.2.7 Logistics

Logistically, both inbound and outbound steel transportation must be considered. This includes freight from Ho Chi Minh City or to the ports which account for approximately \$190,000 per year.

Table XI: Logistical Cost Breakdown

Category	Annual Cost (\$)
Labor	556,800
Energy	25,000
Maintenance	140,000
Facility Costs	119,000
Packaging	90,000
Administrative Costs	75,000
Logistics	190,000
Inventory	1,350,000
Total	2,545,800

14.3 Profit and Loss Statement

Table XII: Product Units and Revenue Table

Product	Units	Price (\$)	Revenue (\$)
New Stainless-Steel Boxes	600,000	20	12,000,000
Refurbished Boxes	300,000	20	6,000,000
Total	900,000		18,000,000

Cost of Goods Sold

For new units.

$$600,000 * 8.62 = \$5,172,000 \quad (7)$$

$$600,000 * 4.50 = \$2,700,000 \quad (8)$$

$$COGS_{new} = 7,872,000 \quad (9)$$

For refurbished units.

$$300,000 * 3.00 = \$900,000 \quad (10)$$

$$COGS_{refurbished} = \$900,000 \quad (11)$$

Therefore,

$$COGS_{total} = \$8,772,000 \quad (12)$$

Gross Profit

$$Gross Profit = 18,000,000 - 8,772,000 \quad (13)$$

$$\text{Gross Profit} = \$9,228,000 \quad (14)$$

EBITDA

$$\text{EBITDA} = 9,228,000 - 2,545,800 \quad (15)$$

$$\text{EBITDA} = \$6,682,200 \quad (16)$$

Net Operating Profit

$$\text{EBIT} = 6,682,200 - 1,000,000 \quad (17)$$

$$\text{EBIT} = \$5,682,200 \quad (18)$$

Estimated Taxes

$$\text{Tax} = 5,682,200 * 0.2 \quad (19)$$

$$\text{Tax} = \$1,136,440 \quad (20)$$

Net Profit

$$\text{Net Profit} = 5,682,200 - 1,136,440 \quad (21)$$

$$\text{Net Profit} = \$4,545,760 \quad (22)$$

Return on Investment

$$\text{Initial Investment} = \$10,000,000 \quad (23)$$

$$\text{Net Annual Profit} = \$4,545,760 \quad (24)$$

$$\text{Payback Period} = 2.2 \text{ Years} \quad (25)$$

$$\text{ROI} = 45.46\% \quad (26)$$

Net Present Value (NPV)

$$\text{PV Annuity Factor} = \frac{1 - (1 + 0.10)^{-10}}{0.10} \quad (27)$$

$$\text{PV Annuity Factor} = 6.1446 \quad (28)$$

$$\text{PV of Future Cash Flows} = 4,545,760 * 6.1446 \quad (29)$$

$$\text{PV of Future Cash Flows} = \$27,932,000 \quad (30)$$

$$\text{NPV} = 27,932,000 - 9,500,000 \quad (31)$$

$$NPV = \$18,432,000 \quad (32)$$

Since the NPV value is substantially positive, the project is highly valuable, profitable, and economically justifiable over its life cycle.

14.4 Triple Bottom Line Matrix

Table XIII: Triple Bottom Line Matrix

Timeline	Cost (-) or Benefit (+)		
	Financial	Human	Environmental
Land Acquisition	-1,950,000	+390,000	0
Shell Factory Purchase	-490,000	0	+3,500,000
Interior Construction	-770,000	+560,000	+230,000
Machinery and Equipment	-3,489,000	+730,000	+500,000
Energy System	-1,230,000	+112,000	+369,000
Raw Material Inventory	-160,000	+85,000	0
Site Prep	-350,000	+220,000	+470,000
Total	-8,439,000	+2,097,000	+5,069,000

This table shows the triple bottom line matrix in which costs and benefits are given values with respect to financial, human, and environmental factors when comparing to the timeline of the project.

Financially, everything is considered a cost as each step of the timeline is relative to the CAPEX to either acquire or produce a specific thing. Therefore, they are all considered costs. The heaviest cost is the machinery and equipment, which require specialized products, causing an increase in costs.

The key human benefit is representative of human job creation value for local labor the value of local construction jobs and improved workspace design causes great benefit to the local community and local skilled workers. It creates the value of specialized training and highly skilled jobs with reduced accident risks. Working with high-quality equipment and machinery. It generates value out of jobs related to sourcing and handling, and the value of labor and administrative jobs for the setup. These values are determined by the compensation provided to the individuals responsible for these sectors. The creation of a skilled job and vocational training

provides benefits not only to the workers but also to the community around them. The highest benefit comes from working with high-quality machinery and equipment due to specialized training, highly skilled jobs, and reduced accident risks. The purchase of a shell factory provides no human benefit as no labor is required, and no loss of labor is caused as a shell is pre-existing.

The environmental benefits are representative of the benefits this project's decisions create for the environment. values are derived from pre-existing structures and avoiding or reducing waste and natural material usage. Furthermore, it is impacted by leveraging DLIP's green park status. values are derived from carbon dioxide emissions avoided, steel recyclability, low waste/scrap generation from the machinery and renewable power sources. The biggest benefit is derived from the shell factory purchase as a pre-existing structure is purchased, avoiding construction waste from machinery and raw materials. This leverages DLIP's Green Park status. The land acquisition has no effect on the environment as the land was allocated for factory use. The raw material inventory has zero net benefit, as all benefit from purchasing the material is mitigated by the carbon emissions of transportation. Therefore, it is considered to have no benefit.

15.0 Business Case and Strategic Feasibility Analysis

15.1 Feasibility Analysis

Table XIV: Feasibility Assessment

Feasibility Domain	Assessment	Rationale
Technical Feasibility	High	The core processes (deep drawing, annealing, digital printing) are established. The key technical challenge integrating high-efficiency servo presses and Industry 4.0 systems is managed by the specialized engineering team. The \$1,000kWp solar array design is validated by the high solar potential of Ninh Thuan.
Economic Feasibility	High	Confirmed by the financial analysis in Section 14, which demonstrated a Net Present Value (NPV) of \$18.43 million and a rapid Payback Period of 2.24 years. The high profit margin (20 USD selling price vs. 8.85 USD variable cost) ensures strong, sustained profitability.
Legal Feasibility	High	The factory is located within the highly regulated Du Long Industrial Park (DLIP), guaranteeing adherence to

		national and provincial industrial codes. The project is legally insulated by Vietnam's commitment to international agreements like the CPTPP and EVFTA, which provide investment protection.
Operational Feasibility	Moderate	Daily operations are feasible, leveraging Vietnam's skilled labor pool. The primary operational risk is the necessity for intensive, specialized training of the local workforce to manage the complex, high-precision Servo Press and Industry 4.0 equipment.
Environmental Feasibility	High	The project meets the highest standards through its closed-loop steel recycling, near-zero water discharge systems, and the commitment to 80%-100% renewable energy usage. The facility is located in an officially recognized Certified Green Industrial Park (DLIP).

15.2 SWOT Analysis

A SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis identifies the internal and external factors that affect the project's success.

Strengths (Internal)

- Closed loop circularity: A 100% recyclable stainless steel model ensures lower raw material cost reliance and superior environmental performance.
- Higher operational efficiency: Integration of Industry 4.0 systems, VSD compressors and servo presses minimize OPEX and maximizes production yield.
- Financial robustness: Project is under budget and delivers a compelling ROI.
- Renewable Energy Integration: 1000 kWp Solar Array in a Region with the highest solar potential in Vietnam.

Weaknesses (Internal)

- High initial capex: Significant up-front investment required for specialized machinery and the PV system.
- Reliance on skilled labor: Advanced technology requires intensive training for the local work force.
- Single product focus: The business is concentrated on a single consumer product. The steel box, increasing market sensitivity.

Opportunities (External)

- Global export access: zero-to-no-tariff access to major global markets (EU and ASEAN countries) via the EVFTA and CPTPP.
- Growing ESG demand: Increasing consumer and institutional demand globally for verifiably sustainable and plastic-free products.
- Infrastructure improvement: Ongoing completion of the North-South Expressway will drastically reduce logistic costs.
- Tax incentives: Regional benefits in DLIP include tax holidays for up to four years.

Threats (External)

- Third of substitutes: High competition from cheap, established plastic containers and other reusable materials.
- Global commodity price volatility: susceptibility to fluctuations in the price of raw stainless steel.
- Grid variability: Potential for instability in the developing country's power grid.

Discussion

Despite the high initial capex, ROI is expected within the next five years. Skill labor, though it relies on skill labor, DLIP guarantees highly skilled, industrially competent workers. Though only a single product is sold, it is of the highest quality and dominates a market currently sparsely inhibited. Susceptibility to price fluctuations is mitigated by internal recycling. Instability caused by developing countries' power grid is mitigated by the PV and battery system.

15.3 Risk Analysis

Table XV: Risk Analysis Table

Risk Identified	Probability	Impact	Risk Rating	Mitigation Strategy
Supply Chain Disruption	Moderate	High	High	Diversify supplier base (Pomina, OSS, scrap), accelerate internal reverse logistics program.
Failure to Achieve Quality Standards	Moderate	High	High	Implement Industry 4.0 Automated Vision Inspection to catch defects immediately post-forming.
Worker Safety Incidents	Low	High	Moderate	Mandate and enforce safety equipment, provide air-conditioned

				transport, implement integrated smart safety sensors and CCTV.
Budget Overrun (CAPEX)	Low	Moderate	Low	Maintain \$750,000 contingency fund; use prefabricated components where possible.
Technology Obsolescence	Medium	Low	Low	Assign a short NUL (5-7 years) for digital systems, budgeting for software and sensor upgrades.

15.4 Corporate Social Responsibility

The project Corporate Social Responsibility (CSR) strategy goes beyond simple compliance, integrating ethical practices into the core operations.

Stellum Industries commits to rigorous auditing of both new steel coil and collected scrap supplies to prevent the inclusion of material sourced from unethical labor or mining practices. failed labor practices is a non-negotiable commitment to the welfare and respect of the workforce, ensuring full compliance with the Vietnamese labor code and providing robust training in a self-working environment, furthermore, ensuring appropriate compensation to the workers. The factory will actively partner with local vocational institutions to develop specialized training programs, ensuring that job creation translates into long-term skill acquisition and upward mobility for the local community in Ninh Thuan.

16.0 Safety Engineering and Ethical Considerations

16.1 Safety Engineering in Design and Operation

Safety is integrated into the factory design process to eliminate or minimize risks at the source, ensuring the health, safety, and welfare of the public and workers.

Safety in Design (SiD)

The architectural layout utilizes reinforced, moisture-resistant walls to specifically separate high-hazard zones, such as the thermal processing and water systems, from the main production areas. This prevents cross-contamination and contains heat/steam exposure.

The area housing the heavy servo presses utilizes isolated foundations to dampen vibration transfer, protecting both the structural integrity of the building and adjacent precision machinery.

The 1000 kW PV system and battery energy storage system are managed by a power control system and integrated inverters, which include essential balance of system components like fuses, DC/AC switchgear, and protection systems to prevent electrical faults and fires.

The selection of electrostatic micro-lubrication dramatically reduces the volume of hazardous forming fluid used by up to 80%. This minimizes chemical handling and volume of associated hazardous liquid waste, lowering the risk of chemical exposure.

Operational Safety and Emergency Systems

All workers are mandated to wear full personal protective gear, including safety gloves, glasses, masks and full-sleeve clothing. Air-conditioned transport and internal cooling units are provided to mitigate the severe risk of dehydration and heat strokes in Vietnam's humid climate. an integrated smart safety system consisting of network sensors and CCTV is installed to monitor restricted zones and operational parameters in real-time. Standard emergency infrastructure is in place, including clearly marked, unobstructed evacuation routes and mandatory staff training in emergency procedures and first aid.

16.2 Ethical Responsibilities

The ethical framework for Stellum Industry is directly founded upon the Engineers Australia Code of Ethics and Guidelines on Professional Conduct, ensuring professional behavior is guided by core values of integrity, competence, leadership and sustainability.

Demonstrate Integrity

The project is managed with honesty and transparency. The design team commits to providing holistic assessments of feasibility and risk (Section 15) and maintaining transparency in costing and financial projections (Section 14). This ensures that the client and all stakeholders rely on objective and truthful information. All persons, including local workers, are treated with dignity, respect, and fairness, adhering strictly to Vietnamese labor laws regarding wages and hours.

Practice Competently

The engineers are responsible for applying specialized knowledge effectively and diligently. The team applies specialized competence in Industry 4.0, circular economy principles, and material science to deliver a tactically sound and high-efficiency solution. A commitment is made to train the local workforce in complex machinery operations and software management, acting as mentors to transfer knowledge from skill to others. All designs are executed in accordance with legal requirements and commonly accepted engineering standards of the day, including compliance checks with the DLIP's environmental license.

Exercise Leadership

The project upholds the reputation of the engineering profession. The commitment to safety, profitability, and social responsibility ensures that the project is a positive example of the engineering profession's role in sustainable development. The project promotes equitable participation by employing local labor and fostering a non-discriminatory workplace environment.

Promote Sustainability

Design actively seeks to create engineering solutions for a sustainable future. The factory holds paramount the health, safety, and welfare of the community and the environment, incorporating social, cultural, and environmental considerations into every engineering decision.

The core strategy of infinite material circularity and maximized renewable energy penetration directly addresses the need to balance the needs of the present with the needs of future generations, minimizing resource consumption and maximizing the public good. The factory is designed to protect and where possible improve the natural environment of Ninh Thuan.

17.0 Expected Outcomes and Projected Impact

Stellum Industries' Stainless Steel Lunchbox Factory is projected to deliver significant positive impacts across environmental, social, and economic domains, establishing a model for sustainable industrialization in Vietnam.

The primary environmental outcome is the successful implementation of a closed-loop production system resulting in substantial resource conservation and pollution reduction. Through the utilization of the 1000 kWp solar array, the factory is projected to achieve 80-100% of its operational power needs from clean energy, providing a direct offset to CO₂ emissions generated by grid power. Process optimization VSD, RHE provides an additional 25-30% energy efficiency gain. The design ensures near-zero manufacturing waste by making the steel scrap 100% recyclable, which is immediately fed back into the material supply chain. The product provides a durable, reusable alternative that directly reduces the volume of single-use plastic containers contributing to landfill and ocean pollution. The commitment to the reverse osmosis and ion exchange system for greywater reuse ensures minimal freshwater consumption and near zero discharge of wastewater, mitigating the environmental impact typical of industrial cleaning processes.

The project will generate long-term benefits for the local community and workforce in the Ninh Thuan province. The factory will generate stable, high-quality employment opportunities. The investment in specialized equipment necessitates vocational training programs leading to the creation of a skilled labor workforce proficient in Industry 4.0 technology, advanced manufacturing, and maintenance. Strict adherence to the Vietnamese Labor Code and explicit commitment to the EA Australia code of ethics ensures a safe, ethical, and equitable working environment with welfare enhancements such as air-conditioned facilities and subsidized transport. The product's success is intrinsically linked to public education, encouraging consumers to adopt reusable lunchboxes, thereby fostering a culture of responsible consumption in both domestic and export markets.

The project is highly valuable, demonstrating that profitability and sustainability are complementary goals. The conservative financial analysis in Section 14 confirms a strong economic case, projecting a net present value of approximately \$18.43 million over 10 years and a rapid payback period of approximately 2.2 years. The factory is projected to generate an annual revenue of \$12 million based on 600,000 unit production run, contributing significantly to regional economic output and export earnings. Investment in self-sufficient systems, such as PV and closed-loop recycling, provides a strong hedge against future volatility in utility and raw material prices, ensuring predictable and stable operation expenditure over the long term.

Stella Industries provides a template for sustainable industrialization by creating a virtuous cycle. Environmental responsibility lowers operational expenditure, which in turn increases profitability and competitiveness. By leveraging Vietnam's favorable trade agreements and the region's abundant renewable resources, the factory is positioned not only for long-term survival but as a future global leader in material circularity, validating the design's commitment to both economic development and environmental stewardship.

18.0 Conclusion

Stellum Industry is a financially robust project with an intended CAPEX of \$9.5M successfully meeting the \$10M budget constraint. This project highlights the financial viability of industrialization in developing countries, evidenced by the \$18.43M net present value and the rapid 2.2-year period back period. Vietnam's Du Long Industrial Park is the optimal site due to its high solar potential and critical access to global markets via the EVFTA and CPTPP, as well as extensive and growing industrial infrastructure promoting foreign investments and industrial growth. Stellum Industries fulfills the client's goal of building a profitable, sustainable, and innovative factory of sustainable goods in a developing economy.

Stellum Industry is a combination of contemporary innovation and sustainability. The core innovation is based upon the infinite recyclability of stainless steel, which creates a perpetual closed-loop production system that eliminates resource degradation. This allows for sustainability and profitability through refurbishing existing products. Furthermore, with the integration of smart systems such as electromechanical servo presses and VSD compressors, which reduce energy consumption and maximize production efficiency. The 1,000 kW/P solar photovoltaic system and BESS aim to power up to 100% of operations, serving as the single largest environmental mitigation measure.

Stellum's technology and architecture are proven through extensive research and past industrial endeavors. The specialized zonal layout and I-shaped production flow highlights technical and operational readiness through robust safety measures in design, PPE, and welfare provisions. Though initial expenditures and reliance on skilled labor carry risks, the youth of the project, Stellum Industries, can mitigate the risks to provide a sustainable and profitable solution. the factory's operation aligns with Engineers Australia's code of ethics, prioritizing the welfare of

workers and the environment. it promotes competence and meaningful leadership to grow sustainable and successful endeavors.

Stellum Industries provides a replicable model for future industrial development, proving that advanced engineering can successfully drive economic growth in a developing nation while adhering to global climate goals. This factory positions the client as a leader in verifiable material circularity and sustainable manufacturing for the global export market.

References

- [1] G. Müller, “Unlocking Opportunities: Sustainable Industrialization in LLDCs,” United Nations Think Pieces on Landlocked Countries, United Nations, May 22, 2025. [Online]. Available: <https://www.un.org/en/landlocked/think-pieces/unlocking-opportunities-sustainable>. [Accessed: Oct. 17, 2025].
- [2] “The Sustainable Development Goals,” United Nations. [Online]. Available: <https://sdgs.un.org/goals>. [Accessed: Oct. 17, 2025].
- [3] “World Steel Association,” worldsteel.org. [Online]. Available: <https://worldsteel.org>. [Accessed: Oct. 17, 2025].
- [4] L. D. D. Harvey, “Iron and steel recycling: Review, conceptual model, irreducible mining requirements, and energy implications,” *Renewable and Sustainable Energy Reviews*, vol. 138, p. 110553, Mar. 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1364032120308376>.
- [5] P. M. Coelho, B. Corona, R. ten Klooster, and E. Worrell, “Sustainability of reusable packaging — Current situation and trends,” *Resources, Conservation & Recycling: X*, vol. 6, Art. no. 100037, Apr. 2020. doi: 10.1016/j.rcrx.2020.100037.
- [6] “Ministry of Natural Resources and Environment,” Vietnam Ministry of Natural Resources and Environment. [Online]. Available: <https://en.monre.gov.vn/>. [Accessed: Oct. 17, 2025].
- [7] “Unilever Vietnam helps suppliers accelerate green transition in supply chains,” *Vietnam News*, Feb. 26, 2025. [Online]. Available: <https://vietnamnews.vn/economy/1692874/unilever-vietnam-helps-suppliers-accelerate-green-transition-in-supply-chains.html>. [Accessed: Oct. 17, 2025].
- [8] “International Energy Agency,” IEA. [Online]. Available: <https://www.iea.org>. [Accessed: Oct. 17, 2025].
- [9] M. Bouyzem and N. Jedlane, “Sustainable development in the automotive industry: A case study of Renault Morocco,” *Revue d’Études en Management et Finance d’Organisation*, vol. 9, no. 2, Dec. 2024. [Online]. Available: https://www.researchgate.net/publication/389601029_SUSTAINABLE_DEVELOPMENT_I

N_THE_AUTOMOTIVE_INDUSTRY_A_CASE_STUDY_OF_RENAULT_MOROCCO_D
EVELOPPEMENT_DURABLE_DANS_L'INDUSTRIE_AUTOMOBILE_UNE_ETUDE_D
E_CAS_DE_RENAULT_MAROC_Meriem_BOUYZEM_Enseignant_cherche. [Accessed:
Oct. 17, 2025].

- [10] E. Barbulescu, M. Nassereddine, and H. S. Rajamani, “Bifacial PV Panels for Green Roofs in Dusty Regions,” in 2024 International Conference on Electrical, Computer and Energy Technologies (ICECET), Sydney, Australia, 2024, pp. 1–6, doi: 10.1109/ICECET61485.2024.10698219.
- [11] “Vietnam-EU Trade: EVFTA Comes Into Effect,” Vietnam Briefing, Aug. 3, 2020. [Online]. Available: <https://www.vietnam-briefing.com/news/vietnam-eu-trade-evfta-comes-into-effect.html/>. [Accessed: Oct. 17, 2025].
- [12] “Viet Nam,” GitHub.org. [Online]. Available: <https://www.github.org/countries/viet-nam/>. [Accessed: Oct. 17, 2025].
- [13] “Why Vietnam’s Infrastructure is Crucial for Economic Growth,” Vietnam Briefing. [Online]. Available: <https://www.vietnam-briefing.com/news/why-vietnams-infrastructure-crucial-for-economic-growth.html/>. [Accessed: Oct. 17, 2025].
- [14] “Decision No. 12/QĐ-TTg amending the planning for the road network for the period of 2021-2030,” Thu Vien Phap Luat (Vietnam Law Library). [Online]. Available: <https://thuvienphapluat.vn/van-ban/EN/Xay-dung-Do-thi/Decision-12-QĐ-TTg-2025-amendment-to-the-planning-for-road-network-for-the-period-of-2021-2030/649962/tieng-anh.aspx>. [Accessed: Oct. 17, 2025].
- [15] “Vietnam’s Solar and Wind Power Success Provides Key Lessons for Other ASEAN Markets,” REGlobal. [Online]. Available: <https://reglobal.org/vietnams-solar-and-wind-power-success-provides-key-lessons-for-other-asean-markets/>. [Accessed: Oct. 17, 2025].
- [16] “Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) - Consolidated Text,” Institute of International Law and Justice (IILJ). [Online]. Available: <https://www.iilj.org/wp-content/uploads/2018/03/CPTPP-consolidated.pdf>. [Accessed: Oct. 17, 2025].

- [17] “Vietnam's Economic Transformation and Trade Integration,” The Australian National University Open Research Repository. [Online]. Available: <https://openresearch-repository.anu.edu.au/items/3392bee1-1e55-4900-aaef-d4ceb4944f5b>. [Accessed: Oct. 17, 2025].
- [18] “Vietnam's Industrial Zones: 2025-2030 Growth Outlook,” Vietnam Briefing. [Online]. Available: <https://www.vietnam-briefing.com/news/vietnams-industrial-zones-2025-2030-growth-outlook.html/>. [Accessed: Oct. 17, 2025].
- [19] “Sustainable Development,” Du Long Industrial Park. [Online]. Available: <https://dulongip.vn/en/sustainability>. [Accessed: Oct. 17, 2025].
- [20] Engineers Australia, Code of Ethics and Guidelines on Professional Conduct, Aug. 2022. [Online]. Available: <https://www.engineersaustralia.org.au/sites/default/files/2022-08/code-ethics-guidelines-professional-conduct-2022.pdf>. [Accessed: Oct. 17, 2025].
- [21] J. Wood, “Generation Z cares more about sustainable buying decisions than brand names,” World Economic Forum, Mar. 18, 2022. [Online]. Available: <https://www.weforum.org/agenda/2022/03/generation-z-sustainability-lifestyle-buying-decisions/>. [Accessed: Oct. 17, 2025].
- [22] “Daylight Sensors Lighting Study,” LEDCity. [Online]. Available: <https://ledcity.io/en/daylight-sensors-lighting-study/>. [Accessed: Oct. 17, 2025].
- [23] G. Kemper, “Solar Panel Angle Calculator,” Solartap. [Online]. Available: <https://www.solartap.com/blogs/solar-101/solar-panel-angle-calculator>. [Accessed: Oct. 17, 2025].
- [24] “What is industry 4.0?” IBM. [Online]. Available: <https://www.ibm.com/think/topics/industry-4-0>. [Accessed: Oct. 17, 2025].
- [25] “What is a variable speed drive?” ABB. [Online]. Available: <https://www.abb.com/global/en/areas/motion/drives/expertise-technology/what-is-a-variable-speed-drive>. [Accessed: Oct. 17, 2025].
- [26] M. A. Arshad, M. Y. Ali, and M. A. Arshad, “Energy, exergy and economic analysis of an annealing furnace,” University of Malaya Repository. [Online]. Available:

https://eprints.um.edu.my/6707/1/Energy%2C_exergy_and_economic_analysis_of_an_annealing_furnace.pdf. [Accessed: Oct. 17, 2025].

- [27] A. M. Almutairi, A. B. Alodan, and M. A. Alsuwailem, “Energy efficiency and waste heat recovery in industrial processes: A comprehensive review,” *Journal of Cleaner Production*, vol. 315, p. 128121, Sep. 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2212827121001578>. [Accessed: Oct. 17, 2025].
- [28] “Comprehensive Guide to Servo Press,” Shuntec Press. [Online]. Available: <https://www.shuntecpress.com/comprehensive-guide-to-servo-press/>. [Accessed: Oct. 17, 2025].
- [29] “When to Choose Variable Speed Drive or Fixed Speed Air Compressors,” *Fluid-Air Dynamics*. [Online]. Available: <https://fluidairedynamics.com/blogs/articles/when-to-choose-variable-speed-drive-or-fixed-speed-air-compressors>. [Accessed: Oct. 17, 2025].
- [30] “10 tháng đầu năm, khối lượng hàng hóa thông qua cảng biển tăng trở lại,” *Báo Vietnam.vn*. [Online]. Available: <https://www.vietnam.vn/10-thang-dau-nam-khoi-luong-hang-hoa-thong-qua-cang-bien-tang-tro-lai/>. [Accessed: Oct. 17, 2025].

Appendices

Appendix 6

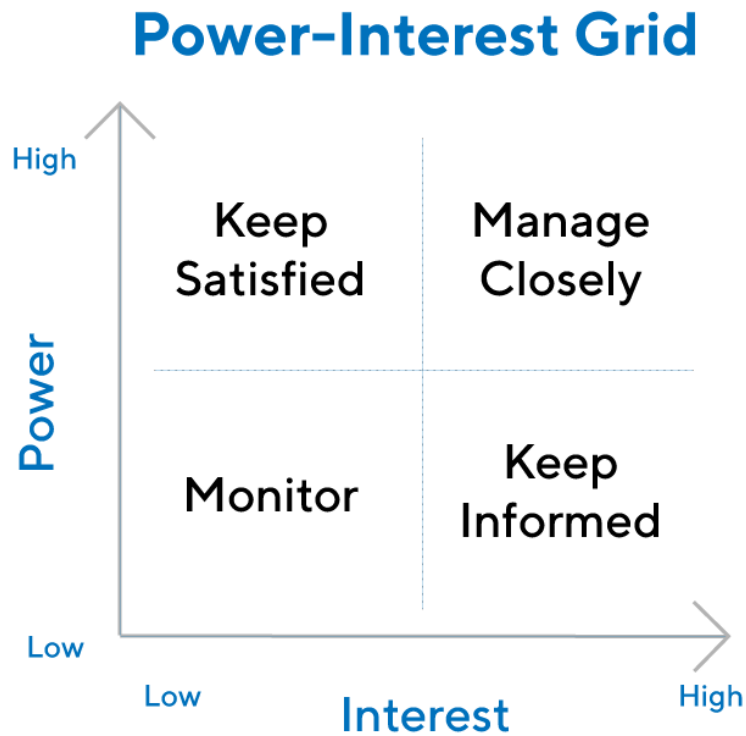


Figure 9: Power-Interest Matrix

Appendix 9



Figure 10: Stellum Industries Website QR code

Appendix 14

Equipment ID	Equipment	Qty	Target Budget (USD)	Selected Model/Spec	Est. Cost (USD)
1.1	Overhead Crane (5 ton)	2	\$60,000	Konecranes CXT Single Girder (VFD)	\$50,000
1.2	Coil Storage Racks	-	\$20,000	Structural I-Beam Coil Cradles (Heavy-Duty)	\$18,000
1.3	Decoiler + Straightener	1	\$40,000	Upgrade: Separate Decoiler & Precision Leveler	\$53,000
1.4	Sheet Cutting Shear / Slitter	1	\$60,000	CNC Guillotine Shear (e.g., HARSLE QC11K)	\$58,000
2.1	Hydraulic Press (400 ton)	2	\$300,000	Upgrade: High-Efficiency SERVO Press (e.g., AIDA/Komatsu)	\$700,000
2.2	Hydraulic Press (250 ton)	2	\$240,000	High-Speed Hydraulic Press w/ Proportional Valves	\$240,000
2.3	Trimming Press (automatic)	2	\$100,000	High-Speed Mechanical C-Frame Press	\$100,000
2.4	Lubrication + Blank Feeder System	1	\$80,000	Automatic Dual-Head Electrostatic Oiling System	\$80,000
2.5	Tooling (dies, molds, punches)	Set	\$180,000	High-Durability Carbide Tooling (coated)	\$180,000
2.6	Conveyor Transfer System	1	\$50,000	Modular Magnetic Belt Conveyor System	\$50,000
2.7	Scrap Collector / Chute	1	\$20,000	Automated Baling/Compacting System	\$50,000
3.1	Electric Continuous Annealing Furn.	1	\$400,000	Upgrade: Recuperative Roller Hearth Furnace	\$450,000
3.2	Ultrasonic Cleaning Line	1	\$80,000	Multi-Stage Immersion with Filtration (closed-loop)	\$80,000
3.3	Electro-polishing Machine	1	\$120,000	Programmable PLC-Controlled Batch System	\$120,000
3.4	Drying Tunnel	1	\$50,000	Forced-Air, Infrared (IR) Drying Tunnel	\$50,000
3.5	Water Filtration & Recycling System	1	\$100,000	Reverse Osmosis (RO) & Ion Exchange (IX) System	\$100,000
4.1	UV Printing Machine (multicolor)	1	\$200,000	Upgrade: High-Speed Digital UV-LED Printer	\$250,000
4.2	Automatic Coating Line	1	\$150,000	High-Precision Spray/Roll Coater (Closed-System)	\$150,000
4.3	Curing Oven (electric)	1	\$80,000	Curing Oven (for base/lacquer coat only)	\$50,000
4.4	Quality Inspection & Vision System	1	\$70,000	High-Resolution Automated Vision System	\$70,000
5.1	Seam Welding/Seaming Machine	2	\$120,000	High-Speed CNC Seamers	\$120,000
5.2	Automated Labeling Line	1	\$60,000	Smart Labeling System with Variable Data Printing	\$60,000
5.3	Leak & Pressure Tester	1	\$40,000	Non-Destructive Air Leak Tester (High-Sensitivity)	\$40,000
5.4	Final Packaging Line	1	\$80,000	Automated Cartoner for Recycled/Compostable Media	\$80,000
5.5	Barcode + Data Logging System	1	\$50,000	ERP Integration with RFID/2D Barcode Track & Trace	\$50,000
6.1	Solar PV Array	-		1,000 kW (1 MW) of N-Type TOPCon Panels, Racking, and Mounting (excluding Inverters/BO	\$650,000
6.2	Battery Energy Storage	-		500 kWh LiFePO ₄ (LFP) Battery Modules, Enclosure, and BMS.	\$150,000
6.3	Inverters & PCS	-		1 MW Hybrid Inverters (manage PV, Battery, Grid, and load), Power Control System (PCS).	\$150,000
6.4	Balance of System (BOS)	-		Cabling, DC/AC Switchgear, Protection Systems (fuses, disconnects).	\$50,000
6.5	Implementation & Software	-		Installation Labor, Permitting, Grid Interconnection, SCADA, and EMS (incl. Daylight Sensor I	\$100,000
7.1	Air Compressors (2-stage, oil-free)	2	\$60,000	Upgrade: VSD Oil-Free Screw Compressors	\$100,000
7.2	Cooling Water System + Pumps	1	\$50,000	Closed-Loop Chiller System with VFD Pumps	\$50,000
7.3	Maintenance Tools & Workstations	-	\$40,000	Standardized Tool Sets & Hydraulic Bench	\$20,000
7.4	Fire Safety, Sensors, and CCTV	-	\$40,000	Integrated IoT/Smart Safety System	\$40,000
7.5	IT & ERP System	-	\$30,000	Cloud-Based ERP/MES System (Subscription-Based)	\$30,000
Subtotal			\$4,070,000		\$4,589,000

Figure 11: Machine Cost Breakdown